

UNIVERSITI TEKNOLOGI MARA

**CHEST X-RAY DIAGNOSIS SYSTEM USING VISION-
LANGUAGE MODELS IN PICTURE ARCHIVING AND
COMMUNICATION**

MUHAMMAD NURHILMAN BIN MOHD ROZALEE

**BACHELOR OF INFORMATION SYSTEMS(HONS.) INTELLIGENT
SYSTEMS ENGINEERING**

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**Chest X-Ray Diagnosis System Using Vision-Language
Models in Picture Archiving And Communication**

Muhammad Nurhilman Bin Mohd Rozalee

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SUPERVISOR APPROVAL

Chest X-Ray Diagnosis System Using Vision-Language Models in Picture Archiving And Communication

By

MUHAMMAD NURHILMAN BIN MOHD ROZALEE
2024814584

This thesis was prepared under the supervision of the project supervisor, Farah Syazwani Mohamed Rashid. It was submitted to the Faculty of Computer and Mathematical Sciences and was accepted in partial fulfilment of the requirement for the degree of Bachelor of Information Systems (Hons.) Intelligent Systems Engineering.

Approved by

.....
Farah Syazwani Mohamed Rashid
Project Supervisor

JULY 9, 2026

STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practice of the discipline.

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MUHAMMAD NURHILMAN BIN MOHD ROZALEE
2024814584

JULY 9, 2026

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ABSTRACT

Medical imaging, specifically chest X-ray (CXR) interpretation, is a cornerstone of clinical diagnosis, yet the process is often hampered by high radiologist workloads and the potential for diagnostic inconsistencies. Current artificial intelligence (AI) implementations frequently provide "black-box" disease labels that lack the explanatory depth required for clinical trust and are rarely integrated into unified Picture Archiving and Communication System (PACS) workflows. This project addresses these gaps by developing an integrated, AI-assisted PACS-style platform designed to transform raw radiographs into descriptive, explainable, and clinically meaningful narratives. The system utilizes a multimodal inference pipeline that pairs the BiomedCLIP vision-language foundation model for discriminative image analysis with the Qwen2.5-1.5B-Instruct large language model for structured report generation. Developed using a Streamlit-based frontend and FastAPI backend, the prototype enables a seamless workflow including image upload, visualization with windowing controls, AI-assisted prediction, and PDF report export. Evaluation was performed using a representative subset of the NIH ChestX-ray14 dataset to assess technical functionality and performance. The findings demonstrate that the system successfully identifies thoracic abnormalities with an average processing time of 16.05 seconds per study. Notably, the pipeline maintained absolute output consistency and provided visual interpretability through Grad-CAM heatmaps. These results signify that integrating vision-language and large language models into a unified radiology environment can significantly reduce cognitive fatigue, improve reporting efficiency, and support medical education. Future work will involve domain-specific fine-tuning of the foundation models and extensive clinical validation by certified radiologists to ensure real-world diagnostic reliability.

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LIST OF ABBREVIATION

CXR	Chest X-Ray
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
ROGUE	Recall-Oriented Understudy for Gisting Evaluation
BLEU	Bilingual Evaluation Understudy
PACS	Picture Archiving Communication System
VLM	Vision Language Model
LLM	Large Language Model
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
AI	Artificial Intelligence
ViT	Vision Transformer
SWIN	Swin Transformer
BART	Bidirectional encoding and autoregressive decoding

X-Ray-CoT	Chest X-Ray Chain-of-Thought
XMI	Extensible Markup Language
Qwen2.5-1.5B-Instruct	Qianwen 2.5 - 1.5 Billion parameter - Instruction-tuned model
LoRA	Low-Rank Adaptation
PCP	Commonly Primary Care Physician
DICOM	Digital Imaging and Communications in Medicine
RIS	Radiology Information System
HL7	Health Level Seven
NIH	National Institutes of Health
BiomedCLIP	Biomedical Contrastive Language–Image Pretraining
OHIF	Open Health Imaging Foundation
Grad-CAM	Gradient-weighted Class Activation Mapping

CHAPTER 1

INTRODUCTION

This chapter provides the background and rationale for the study. It also gives detail of background study, the issue and problem that led to research such research question, research objectives, scopes, significant and expected outcome.

1.1 Background Study

Medical imaging has a significant role to play in clinical diagnosis and treatment planning in the era of globalization. The chest X-ray (CXR) is one of the most commonly used diagnostic systems to identify diseases like pneumonia, tuberculosis, and lung cancer, among others. The World Health Organization (WHO, 2023) reports that in the world, over 2 billion chest X-ray tests are done annually, which emphasizes the importance of radiology in health care systems of the population. A chest X-ray gives a two-dimensional image of the thoracic area where clinicians can determine the condition of the lungs and other diseases. The variation in tissue density is represented by the variation of shades of gray, and it is possible to distinguish anatomical structures and possible diseases (Imaging the Chest, 2001). Although it was used widely and has a diagnostic value, the rising number of chest X-ray tests has added pressure to radiologists.

A number of studies have observed that workloads and extended interpretation periods can be a cause of radiologist fatigue and inconsistencies in diagnostic errors. Stec et al. (2018) established that lengthy reading can result in diagnostic error, especially when the subtle findings exist. In the same way, Hopstaken et al. (1996) stressed that in order to diagnose pneumonia based on chest X-ray images, one should closely follow up. These difficulties give rise to the necessity of tools that will be able to help

radiologists and enhance the uniformity and decreasing the time spent on interpreting the results.

The Picture Archiving and Communication System (PACS) has been embraced by many healthcare institutions to handle an extensive number of medical images. PACS allows effective storage, retrieval, and viewing of medical images within hospital departments and has become an essential element of the current radiology practices (Arora & Mehta, 2014). Artificial intelligence (AI) is a supportive technology in the field of medical imaging that has been receiving attention in recent years. Past studies have demonstrated that AI-based solutions, which are Vision Language Model (VLM) and Large Language Model (LLM), can help discover the abnormalities and give explainable output and also place the most important cases in the focus of the radiologists.

However, although the current studies shows the potential of AI in interpreting chest X-rays and the significance of PACS in the radiology process, there is a lack of focus on the implementation of AI-aided interpretation into a PACS-like environment that recreates the real-world clinical practice. Also, there is no strong focus on offering interpretable and clear outputs that aid radiologists in the diagnostic process. The study will fill these gaps by showing an AI-enhanced diagnostic workflow in a simulated PACS setting, with a greater focus on assisting radiologists by enhancing their consistency, efficiency, and workflow integration.

1.2 Problem Statement

Manual reading of chest X-ray is a very slow process requiring extensive experience and consistency, particularly for subtle thoracic abnormalities. With the increasing volume of imaging exams, radiologists risk increased cognitive load and a possibility of less efficient and consistent reporting. There are numerous approaches to AI-assisted chest X-ray interpretation, most of which have primarily focused on disease prediction with little explanation of their outputs. The black-box nature of these predictions reduces transparency of the AI-driven diagnosis and diminishes the trust that clinicians place in them. There is thus a need for an AI-aided interpretation system that integrates the Vision-Language Model and Large Language Model, to generate descriptive, explainable and clinically meaningful radiology reports instead of prediction only.(He et al., 2025)

Medical images can easily be stored and retrieved with the help of Picture Archiving and Communication Systems (PACS), though this has limited usefulness in decision support during medical image interpretation. Current AI-powered chest X-ray systems are predominantly developed as isolated research prototypes, and typically perform individual tasks like disease prediction or summarization of the reports. In such separate implementations, image upload, AI prediction, report production and result management remain scattered, decreasing usability and the level of AI usage in radiology. It is therefore essential to have an integrated PACS-style platform that brings image upload, visualization, AI-assisted disease prediction, report generation, study management, and report export together in one workflow to improve efficiency and user experience.(Pham et al., 2025; Thawakar et al., 2024)

While studies over the last few years have seen encouraging predictive results with Vision-Language Model and Large Language Model, there is less research into the overall performance of integrated AI-powered medical imaging systems in operation. Existing literature is mainly concerned with prediction accuracy and provides little insight on functionality, response time, workflow integration, system stability and

practical usability in a PACS-style setting. As a result, how practical it is to deploy such systems in real radiology workflows is still not well studied. A thorough evaluation is therefore required to determine the performance and reliability of the whole integrated system and the AI models themselves.(Joy, 2024; Y. Wang et al., 2025)

1.3 Research Question

Three research questions have been identified prior to conducting this research:

1. How effectively can a pretrained Vision-Language Model and a pretrained Large Language Model be combined to generate descriptive, explainable, and clinically meaningful radiology reports from chest X-ray images, rather than disease predictions alone?
2. How can image upload, visualization, AI-assisted disease prediction, report generation, study management, and report export be integrated into a single PACS-style workflow to improve usability and operational efficiency?
3. How does the complete integrated system perform in terms of technical functionality, response time, output consistency, system stability, and workflow integration within a PACS-style environment?

1.4 Research Objectives

Three research objectives need to be accomplished for this research:

1. To implement a pretrained Vision-Language Model together with a pretrained Large Language Model for generating descriptive, explainable, and clinically meaningful radiology reports from chest X-ray images within a simulated PACS workflow.
2. To develop an integrated PACS-style platform that supports chest X-ray upload, image visualization, AI-assisted disease prediction, report generation, study management, and report export within a single workflow.
3. To evaluate the complete integrated system in terms of technical functionality, response time, output consistency, system stability, and workflow integration within a simulated PACS-style environment.

1.5 Research Scope

The research scope of this project is focused on developing the prototype PACS-integrated chest X-ray diagnosis system by applying the pretrained Vision-Language Model (VLM), which is BiomedCLIP, to automate the process of the interpretation of medical images and the pretrained Large Language Model (LLM), which is Qwen2.5-1.5B-Instruct, for descriptive output from Hugging Face. The scope of this is limited to the 2D frontal chest X-ray image that has been received from Kaggle, which contains public datasets. The datasets contain at least 110k X-ray images that will be used for demonstration purposes only for the system. The system workflow will enable the user to upload X-ray images, which will undergo processing, after which they will be interpreted and analysed using pre-trained VLM and then with the assistance of LLM, the disease would be identified. Technical functionality of the system (stability of system that will not crash when images are uploaded, etc.) will be examined during the project evaluation. Is the image fed into the system's pretrained model without error and output obtained without error? Then, assess system performance such as response time in seconds, and the consistency of the system output, including the same output after uploading the same CXR image, comparing the existing AI integrated with the PACS environment, including uploading images, image display, performance of AI text generation, and report output.

1.6 Research Significance

The research significance purpose of this study is to develop a prototype Chest X-ray (CXR) that integrates a pretrained Vision Language Model (VLM) and a pretrained Large Language Model (LLM) into a simulated PACS workflow. The function of the system is basically to automatically analyse the CXR image and give explainable output to assist the radiologist and medical worker in producing consistent output that is faster than they can interpret by themselves. This research contributes to the medical imaging and AI field by exactly presenting how explainable vision-

language models and large language models can be applied in a PACS workflow environment for assisting in clinical decisions. This study focuses on the feasibility of using a public dataset rather than relying on a private patient dataset. The benefits of this research are basically for the radiologist and medical worker because the prototype can analyse and generate explainable output that is consistent and faster, which can reduce the cognitive workload. Healthcare also benefits because the prototype can improve workflow efficiency and reduce the reporting time. This also benefits the patient indirectly due to the faster diagnoses and lower human error. Finally, the prototype can benefit the medical student and trainee radiologist because it can be a platform that supports learning and diagnostic reasoning by combining image features with clinical explanations.

The research also supports the Sustainable Development Goal (SDGs), which is SDG 3, Good Health and Well-Being, where the prototype enables faster interpretation than manual, which is more consistent with explainable diagnostics. The system supports contributing to improved healthcare system performance, reducing diagnostic error, and better patient solutions. This also supporting the SDG 4 which is Quality Education that ensure the prototype with natural language explanation of the X-ray findings will serve the prototype as the support mechanism for the learning tool for medical trainees that profiting in enhance diagnostic skills and ensure self-directed learning. In particular, this tool has the potential to support under-resourced medical facilities and rural clinics where access to trained radiologists is limited or unavailable. By providing rapid, automated, and explainable preliminary chest X-ray interpretation, the system can help reduce diagnostic delays in such settings, directly reinforcing SDG 3 (Good Health and Well-Being) through more equitable access to diagnostic support.

1.7 Expected Outcomes

The expected output for this project is a list of descriptive and interpretable textual output that has been generated by a pretrained Vision-Language Model (VLM) and a pretrained Large Language Model based on the chest x-ray images, presenting the

ability in foundation models for automated CXR interpretation. The model used is only for inference components, excluding training, to provide a medically meaningful explanation of identified abnormalities.

A simulated-style interface will be developed to assist in prototype demonstration and visualization. The interface will allow the user to upload an image, display the image for viewing or monitoring, and generate explanation output side by side to represent how the model outputs could be integrated into a radiology workflow. The interface's purpose is to demonstrate the PACS environment, like that used by hospitals, to demonstrate integration and visualization.

The system will be evaluated based on functional testing, system performance, and integration within the simulated PACS workflow. The technical functionality will evaluate if the image is successfully being processed by the pretrained model, producing output without error, and the stability of the system, such as not crashing when uploading an image. Next is system performance, which is response time per second in generating output and consistent output generation with the same input. Finally, the workflow integration will be evaluated based on if generated output is present, enabling coherent interaction between image monitoring and also AI-generated interpretation.

CHAPTER 2

LITERATURE REVIEW

A literature review is a critical assessment of existing published research studies, articles, and academic sources on a given topic of study used to comprehend what is known, what constraints exist, and why new research is necessary.

2.1 Medical Image

Medical imaging has seen a huge increase in workload, and radiology departments are finding themselves under strain, with systems struggling in the diagnostic workflow. The world has seen a gap in radiologists and an increase in imaging studies, resulting in increased workload, delayed report turnaround, and diagnostic mistakes, states (Tanno et al., 2024) Interpretation of images should be matched with the time taken for manual report writing; radiologists must overcome the tension between these two to maintain efficiency and avoid cognitive fatigue.

Also, there is significant inter-observer variation in the radiological interpretation of the same image with different clinicians arriving at different interpretations. Both human-written and AI-generated reports can be inaccurate, as noted by (Tanno et al., 2024), requiring the use of assistive systems instead of fully automated ones. In conclusion, these hurdles illustrate not just the limitations of interpreting images, but also the reporting and documentation requirements that drive radiology workflows, and is where AI-powered systems can come into play and help the clinician, lower workload, increase consistency and ensure human oversight.

2.1.1 Chest X-Ray

CXR, known as chest X-ray imaging, has been the most common anatomy and pathology examination that has been performed radiologically. Based on the X. Wang et al. (2017) statistic, almost 2 billion CXRs have been

performed every year because of the cheap cost, high demand, and ability to identify thoracic diseases such as pneumonia, tuberculosis, cardiomegaly, pulmonary edema, and lung cancer. Even while CXRs are clinically significant, they can be difficult to interpret because of the slight differences in grayscale intensity, which might result in overlapping anatomical structures that could influence the harmful outcome. According to X. Wang et al. (2017) even for deep learning with strong training on big, large datasets, illness detection and recognition have become crucial tasks. Difficult inter-patient variability is a component of integrating; the noise of the pictures' variability, the images, and the existence of comorbidities are all components of integrating. Due to these difficulties, CXR interpretation has become the primary issue in the clinic challenges, particularly when it comes to the pool of CXR images that need to be analysed and published.



Figure 2.1 shows an example of a CXR image from the NIH Chest X-ray Dataset
(Source: Kaggle, 2025).

2.2 Clinical Challenges

The rapid growth in medical imaging volume has placed significant pressure on radiology departments, creating systemic bottlenecks in diagnostic workflows. According to (Tanno et al., 2024) a global shortage of radiologists combined with

increasing demand for imaging studies has led to heavier workloads, longer report turnaround times, and increased risk of diagnostic error. Radiologists are required to balance image interpretation with time-consuming manual report writing, which reduces efficiency and contributes to cognitive fatigue.

Additionally, substantial inter-observer variability exists in radiological interpretation, where different clinicians may reach different conclusions from the same image. This inconsistency can lead to diagnostic uncertainty, repeated examinations, and delayed clinical decisions. (Tanno et al., 2024) further highlighted that both human-written and AI-generated reports may contain errors, emphasizing the need for assistive rather than fully autonomous systems. Overall, these challenges demonstrate that radiology workflows are constrained not only by image interpretation but also by reporting and documentation demands, highlighting the need for AI-assisted systems that support clinicians, reduce workload, and improve consistency while maintaining human oversight

2.2.1 Diagnostic Error

Radiology reporting is a time-consuming cognitive skill that involves abstracting visual observations into medical language. (Islam et al., 2025) empirical studies have found manual report writing to be a key bottleneck in clinical processes, leading radiologists to spend less time interpreting images and more time on documentation. Free-text reports also result in different choices of terminology and writing style. However, identical findings may be described using different terminology by different clinicians, which makes it more challenging for downstream tasks to perform data mining and quality auditing or provide clinical decision support. (Tanno et al., 2024) noted that the lack of consistency in human reports leads to diagnostic uncertainty and reduces their value as training data for AI systems. Automated reporting aims to reduce these concerns by generating uniform interpretations of findings and impressions. However,

creating clinically accurate and linguistically coherent reports involves modelling not only visual features but also medical language at considerable depth which motivates adoption of multimodal architectures.

2.3 Radiology Workflow

Today's radiology workflow is a complex multimodal chain with the PACS acting as the digital core for storage, access, and visualization of imaging informatics. It is the organized system for handling a variety of medical images (X-rays, CT, and MRI) with radiological reports serving as the official documentation of a clinician's analysis. This process is now under considerable stress with a worldwide shortage of radiologists and extremely high annual imaging data volume leading to diagnostic delay and radiologist overload (Huemann et al., 2024; Islam et al., 2025; Ryu et al., 2025).

Comparative reading shows the essentially human-driven process of reviewing a patient's past images and reports to track disease progression or monitor treatment efficacy remains a fundamental step in this workflow. In interpretation, clinicians have a correct chain of thought ranging from recognizing abnormal visual features to formulating pathophysiology and constructing the final diagnosis. The manual creation of unstructured, textual reports at this stage is one of the main clinical bottlenecks, which can give rise to inter-observer variation and potential diagnostic error (Huemann et al., 2024; Islam et al., 2025; Ryu et al., 2025).

2.3.1 Picture Archiving and Communication System (PACS)

PACS acts as the digital foundation of today's radiology departments, offering centralized storage, retrieval and distribution of medical images such as CT scans, chest x-rays and MRIs. (X. Wang et al., 2017) stated that PACS as a big data volume of imaging knowledge database since it collects large collective quantities of radiological studies including reports with its time and date. This integration provides the opportunity to follow-up

patients longitudinally and allows for a continuity of care where both old image records can be compared with new ones. By the substitution of film archives, PACS brings tremendous benefits in terms of workflow efficiencies and speed on sharing image information among different hospital districts.

However, traditional PACS system was originally designed as a data tracking systems and not an analytical tool. They structure images based on metadata like patient ID, examination dates and modalities, but have zero knowledge of image content. A typical PACS does not interpret visual patterns, identify the pathological findings and draw inferences on clinical grounds from previously stored images. This makes it necessary for radiologists to perform detachable tasks such as detection of abnormalities, urgent case prioritisation and reading summary. This dependency on human interpretation is not scalable, especially in a high-throughput setting where thousands of imaging studies are accumulated daily.

In addition, PACS does not allow for automated detection of anomalies or decision support. It is not able to interpret suspicious areas, suggest possible diagnoses, or create initial reports. No priority given to images at storage level, irrespective of the clinical urgency. As a consequence, workflow is limited by the presence of readers specialized in the field and time delays may be experienced for urgent situations (emergency imaging). In this context, PACS is essentially a passive archive that stores and transmits imaging data but does not participate in the reasoning process for image interpretation.

These limitations expose a serious chasm between the availability and use of data in clinical radiology. PACS provides access to images, but does not turn the raw imaging data into useful clinical information. This has spurred studies that aim to incorporate intelligent systems, such as vision–language

models and large language models, into PACS directly. In addition to the role of PACS being confined to that of merely providing a holding place for medical images, future applications of PACS may benefit from the inclusion of semantic capabilities and automatic interpretation support systems currently lacking.

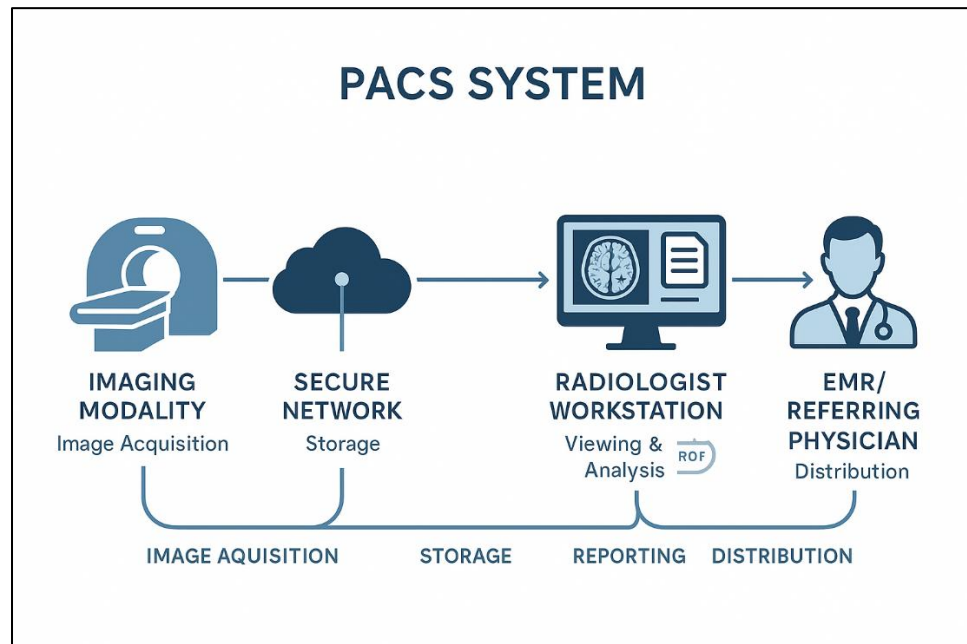


Figure 2.2 shows an example of a PACS system architecture from Curogram
(Source :Amoroso, 2025).



Figure 2.3 shows an example of a PACS system interface
(Source : FUJIFILM Healthcare Americas Corporation, 2025).

2.3.2 System Integration

More recent studies have proposed integrating AI systems into the Picture Archiving and Communication System (PACS) to be able to analyse medical images in real time or close to it upon the capture and storage of an image (Y. Wang et al., 2025). As opposed to the PACS being an archive and retrieval-only service, the proposed approach turns it into a smart processing space whereby image data are automatically analysed once they are stored.

Vision-Language Models (VLM) are the focus of this integration and are used to project chest X-ray images to uniform higher-level semantic features. Such representations contain clinically important visual data, such as anatomy, pathology patterns, and spatial data. These representations are then inputted into Large Language Models (LLM) that give explainable output in the form of structured textual interpretations to assist in bridging the gap between image perception and clinical reporting. Y. Wang et al.

(2025) also suggested governance-ready models that were aimed at safe implementation in clinical workflows. Their model uses an abstention mechanism, which enables the system to make no predictions when there is low confidence or when the inputs are out of the training distribution. This minimizes the chances of giving erroneous results as the proven diagnoses.

Moreover, the framework has a human-in-the-loop design, according to which automated outcomes can be checked and confirmed by clinicians until a final decision is made. This will guarantee that medical professionals will be left in ultimate clinical decision-making even though they will still enjoy the use of computational support. The routine and time-consuming processes can be automated with the incorporation of AI into PACS. These are chest X-ray recognition, indexing and retrieval of disease tagging, and generating of preliminary draft reports. Such processes can be automated to lessen the amount of clerical work and enhance documentation reliability.

Notably, these systems do not mean that radiologists should be replaced. They, instead, act as decision-support tools. Through its capability to ensure clinician supervision and the implementation of clear safety technologies, AI-enhanced PACS allows a more scalable, more efficient, and safer radiology workflow. Figure 2.3 shows the simulation of AI in the PACS environment and shows where the AI platform, such as the vision language model and large language model, can be implemented in the PACS viewer.

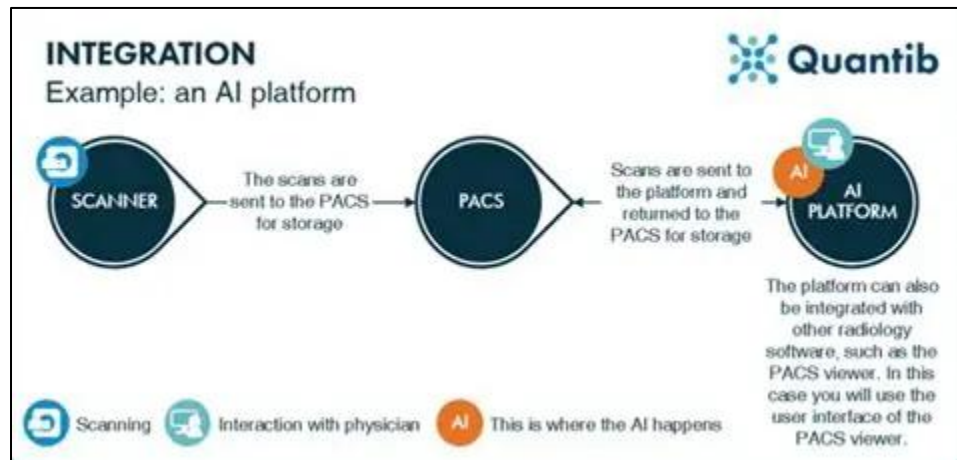


Figure 2.4 shows the explanation of how PACS integration with AI happens in the workflow. (Source : Quantib - DeepHealth.,2025)

2.4 Image Captioning

Medical image captioning is a branch of automated textual image-related textual description generation used to provide a summary of clinically relevant information in radiographic images. In the case of chest X-rays, these involve identifying anatomical structures and identifying pathologies as well as characterizing their locations in accordance with the known medical science. X. Wang et al. (2017) demonstrated that large-scale datasets of chest X-ray images with radiologist-written reports can be used to teach models associations between visual patterns and clinical words under weak supervision, and report generation can be considered a vision-language, not a simple classification, task.

Medical image captioning is more accurate and must be done in line with clinical principles than general image captioning. Created descriptions should be clear, give a description of findings and anatomical location, and should not be ambiguous. Newer methods have thus employed transformer-based vision-language systems that combine visual and language data and generate multi-sentence outputs that are similar to reports written by radiologists and not single labels.

It was shown that transformer-based models are able to produce consistent and semantically relevant radiology reports, such as descriptions of multiple abnormalities observed within a single image (Islam et al., 2025) On the same note, Tanno et al. (2024) discovered that paired chest X-ray images and reports from large-scale vision-language models generated clinically consistent narratives equivalent to standard human-generated reports.

Recent works as well have focused on interpretability in medical image captioning. Ng et al. (2025) demonstrated that semantic grounding is enhanced by including chain-of-thought reasoning, and generated reports are more associated with clinical reasoning. In general, medical image captioning has become an important part of AI-assisted radiology, which helps to use the standardized reporting and decreases the amount of documentation.

2.4.1 Pretrained Vision Language Model

Due to their ability to transfer knowledge from a large image-text database to a specific task in which labeled medical data is limited, pre-trained vision-language models have gained increasing significance in medical image analysis. Training task-specific models from scratch is hard in the field of radiology, where obtaining expert annotations for chest X-rays is expensive and time-consuming (Ryu et al., 2025; Zhang et al., 2025). Pretrained VLM can be used for zero-shot disease classification, image-text retrieval, and evidence localization with only little modification, as it learns representations from paired image-text data (Radford et al., 2021; Zhang et al., 2025).

Most of today's VLM are based on the dual-encoder model, which was laid out in the preliminary work of Radford et al. (2021) the Contrastive Language-Image Pre-training (CLIP) framework. In this design, an image

encoder and a text encoder are trained to embed an image and text description into a common embedding space such that the similarity between the image and text description can be directly computed by cosine similarity. This contrastive setting enables the model to classify an image by referencing class descriptions in natural language rather than a predetermined class set, upon which the zero-shot inference relies. Medical imaging differs from natural images, though, due to a significant shift in the modality of images and the limited amount of data available; general-domain CLIP models trained on natural images do not transfer well to medical imaging (Zhang et al., 2025)

To bridge this domain gap, a number of studies have extended the contrastive vision-language approach to the biomedical domain. One of these is BiomedCLIP (Zhang et al., 2025) which is a biomedical foundation model pretrained on PMC-15M, the PubMed Central database containing fifteen million pairs of figures and their captions from biomedical articles. BiomedCLIP can align features from the radiographic images with biomedical terms using a combination of a Vision Transformer (ViT-B/16) image encoder and a PubMedBERT text encoder. Consequently, it delivers high zero-shot and few-shot results on various biomedical imaging tasks such as chest X-ray classification, and performs better than general-domain CLIP and previous biomedical VLM on retrieval and classification tasks (Zhang et al., 2025) Being a dual-encoder vision-language model, BiomedCLIP has a separate image and text encoder, making it possible to compare a chest X-ray to a set of thoracic disease descriptions and output calibrated similarity scores without any task-specific training.

This ability is directly applicable to this study, for which one is required to predict disease from chest X-rays, but with no access to large labelled training sets or high-performance computing resources, and under an inference-only constraint. Unlike generative encoder-decoder captioning

methods that must be trained on paired image-report data, BiomedCLIP is a discriminative image-text matching system that can be used in a zero-shot approach without the need for image-report pairs for training. Therefore, BiomedCLIP is employed in this study to extract visual features and to generate ranked disease predictions with confidence scores, which are then fed to a large language model to generate the report.

Recent community benchmarks reinforce the suitability of contrastive vision-language models for the recognition of thoracic diseases. For the CXR-LT 2024 challenge, the evaluation extended to 377,110 chest X-rays and 45 disease labels, and an explicit zero-shot generalization challenge for unseen findings was added (Lin et al., 2025). Multimodal image-text models and zero-shot learning strategies were among the best for long-tailed and unseen disease classification, as CLIP-style models such as BiomedCLIP were found to be suitable for the needs of real-world chest X-ray interpretation, which often faces challenges from rare diseases and insufficient labels (Lin et al., 2025).

In addition to classification, BiomedCLIP can be used for explainability using a gradient approach. The model provides a differentiable image-text similarity score, thus it allows using Gradient-weighted Class Activation Mapping (Grad-CAM) proposed by Selvaraju et al. (2017) on the final vision transformer block to derive a heatmap indicating the most important image regions for a particular prediction. This adds to the numerical confidence scores by providing visual interpretability for the predictions, without the addition of any more model to the pipeline (Selvaraju et al., 2017; Zhang et al., 2025).

I. BiomedCLIP

BiomedCLIP is a biomedical vision-language foundation model based on the contrastive dual-encoder architecture of CLIP, adapted to the biomedical domain (Zhang et al., 2025). It consists of two components: a Vision Transformer (ViT-B/16) image encoder which divides the input image into fixed-size patches and models the long-range dependencies using self-attention, and a PubMedBERT text encoder which represents biomedical language. Pretrained on the PMC-15M dataset of fifteen million figure-caption pairs, the two encoders are trained together using a contrastive objective, which ensures that similar image-text pairs fall close together in the embedding space while dissimilar pairs are pushed apart (Zhang et al., 2025). This pre-training in the biomedical domain enables BiomedCLIP to recognize radiographic patterns and map them to clinical terminology far better than general-domain CLIP.

During use, BiomedCLIP performs zero-shot classification by encoding a chest X-ray and a set of textual disease descriptions into the shared space and computing the cosine similarity between the image embedding and each text embedding. These similarities are passed through a softmax function to generate calibrated confidence scores, and the most probable thoracic finding and the most highly ranked alternatives are determined. Because the model exposes a differentiable similarity score, Grad-CAM (Selvaraju et al., 2017) can be applied to its last Vision Transformer block to create a heatmap that shows the regions contributing to each prediction, providing built-in visual explainability without a separate model.

One strength of BiomedCLIP for this study is that it provides medically informative predictions in an inference-only, zero-shot setting, without the need for specific training data or fine-tuning (Zhang et al., 2025). Some limitations must be mentioned too. BiomedCLIP is a discriminative matching model that provides disease predictions and confidence scores instead of full narrative reports, and thus needs to be paired with a language model to generate narrative text. It is also bounded by the set of disease descriptions it is given and by the knowledge it captures during pretraining. These points are the reason for using BiomedCLIP together with a pretrained large language model for downstream report generation.

2.5 Explainable output

Interpretability in medical imaging seeks to generate textual narratives that rationalize and explain predictions of the model as radiologists communicate their diagnostic reasons. Explainable algorithms do more than just make a final decision or label and generate structured interpretations that describe the observed abnormality and its clinical implications. Initial works formulated this problem as a vision–language task that requires mapping visual concepts from medical images into meaningful natural language rationales. This takes disease classification a step even further since it models the descriptive and interpretative reasoning that radiologists use for image-based reporting.

Recently, explainable medical AI has seen significant advancement due to the Transformer-based vision–language architectures with encoder-decoder design. In such systems, visual encoders derive semantic information from image content, and language decoders generate textual explanations conditioned on that rich set of visual cues. Islam et al. (2025) showed that models of this nature are capable of generating fluent and semantically valid descriptions for chest X-ray images, detailing multiple

health issues through coherent medical terms. This ability is further bolstered by large-scale vision-language models that can be trained on paired data of chest X-rays and radiology reports to generate clinically reasonable rationales approximating expert-level reasoning (Tanno et al., 2024).

However, despite these lines of progress, it still remains difficult to create explainable output that is visually and linguistically precise. Models must suppress hallucinations, are required to properly connect visual evidence with anatomy, and should communicate their results in a standardized clinical language. Ng et al. (2025) stressed the necessity of interpretability through reasoning-driven schemes that explicitly reveal intermediate inference steps, a process that precedes reaching ultimate conclusions. In general, explainable output is a key application of large language models in medical imaging and has high potential to boost transparency, clinician trust and decision-making and requires system design with caution for patient safety and clinical reliability.

2.5.1 Pretrained Large Language Model

The pretrained large language model is complementary to AI-assisted radiology, converting structured prediction outputs into coherent, human-readable clinical narratives. While a vision-only classifier produces disease labels and confidence values, an LLM can convert these outputs into descriptive reports that include anatomical context and standard radiological terminology (Ng et al., 2025; Tanno et al., 2024). More recently, reasoning-oriented and instruction-tuned models that follow structured prompts have been found to generate more consistent and controllable text than earlier autoregressive decoders such as GPT-2 (Ng et al., 2025).

Open-weight LLM are particularly appealing for medical applications due to their support for local, reproducible, and privacy-preserving deployment.

Open, publicly released models have demonstrated competitive generative and reasoning performance, and evidence such as X-Ray-CoT (Ng et al., 2025) indicated that interpretable diagnostic reasoning is possible with chain-of-thought prompting of open models. Large models are still resource-intensive, however, which has led to the creation of smaller instruction-tuned models that maintain strong instruction-following ability while remaining efficient on moderate hardware (Qwen Team, 2024; Yang et al., 2024).

Alibaba's Qwen team has developed the Qwen2.5 model family which is a set of open-weight, instruction-tuned transformer language models with a variety of parameter sizes (Qwen Team, 2024; Yang et al., 2024). The large multilingual corpus and instruction following ability make Qwen2.5 well suited for pretraining, while also being aligned with supervised instruction tuning so that it possesses strong instruction-following, structured-output, and reasoning capabilities. The smaller models, such as Qwen2.5-1.5B-Instruct, are designed for lightweight and efficient deployment, providing a good balance between output quality and computational cost (Qwen Team, 2024).

For this study, Qwen2.5-1.5B-Instruct is used as the pretrained language backbone for report generation. This choice is based on three practical reasons. First, the system is inference-only and does not require fine-tuning, so a strong instruction-following model capable of reliably following a structured prompt template is required; Qwen2.5-1.5B-Instruct satisfies this while remaining deterministic under greedy decoding (Qwen Team, 2024). Second, the model's small 1.5-billion-parameter size enables it to run in-process on the available hardware without a dedicated inference server, which is crucial for a prototype designed to demonstrate feasibility rather than to rely on large-scale computing. Third, it is open-weight, which promotes transparency and reproducibility, consistent with the goals of

academic research. Qwen2.5-1.5B-Instruct is therefore chosen because it offers strong instruction-following at a scale well suited to lightweight, inference-only deployment.

The usefulness of Qwen-family models for radiology text tasks has also been tested empirically. Hu et al. (2025) compared the performance of nine large language models, one of them being the Qwen model family, in summarizing radiology report impressions for lung cancer using zero-shot, one-shot, and few-shot prompting. They found that current large language models are able to generate impressions with high completeness and correctness, although conciseness and verisimilitude still require human oversight Hu et al. (2025). This supports the assistive, human-in-the-loop role of the present study and the decision to use an instruction-tuned Qwen model for prompt-driven report generation.

To combine BiomedCLIP and Qwen2.5-1.5B-Instruct into a single inference pipeline, which incorporates a domain-specific vision-language model and an instruction-tuned language model. To achieve this, BiomedCLIP delivers medically informed visual predictions and confidence scores, while Grad-CAM explainability clarifies the predictions into descriptive, radiology-style reports, thanks to the model architecture of Qwen2.5-1.5B-Instruct (Hu et al., 2025; Qwen Team, 2024)

I. Qwen2.5-1.5B-Instruct

The Qwen team of Alibaba has published the Qwen2.5-1.5B-Instruct, an open-weight instruction-tuned large language model (LLM). The Qwen team at Alibaba has released the open-weight Qwen2.5-1.5B-Instruct instruction-tuned large language model (LLM). It is a decoder-only Transformer language model that has been pre-trained on a lot of multilingual text data, followed by a

supervised learning phase to train the model to follow instructions and generate structured outputs and to show strong reasoning capabilities. One of the smaller variants in the family, the 1.5-billion-parameter model is engineered to be lightweight and efficient, delivering a balance between performance and computational resources (Qwen Team, 2024) .

The proposed system's report-generation component is a model named Qwen2.5-1.5B-Instruct. Provided with a structured prompt template with BiomedCLIP's disease predictions and confidence levels, it generates a radiology-style report, which includes sections for Technique, Findings, and Impression. Deterministic (greedy) decoding combined with a repetition penalty is used to achieve consistent and repeatable results for the same input and to maintain the output in the expected format, not conversational responses, as with the case of an iterative decoding, while retraining the model (Qwen Team, 2024).

The key benefit of Qwen2.5-1.5B-Instruct is that it achieves competitive instruction-following and language-generation performance at a smaller scale that can be deployed in-process on modest hardware without the need for a dedicated inference server, which is essential for the inference-only prototype (Qwen Team, 2024; Yang et al., 2024) Its open-weight release further supports transparency and reproducibility. However, like other compact language models, it has weaknesses: because the model does not analyse the chest X-ray image directly, its output quality depends on the accuracy of the upstream BiomedCLIP prediction, and its limited capacity may not allow for as deep a clinical reasoning as larger language models. Such trade-offs are reasonable for a

demonstration-oriented system that prioritises efficient, reproducible, inference-only operation.

2.6 Related Works

The related work of this literature review is referred from existing works and clinical systems that enable benchmarks for comparison performance, result analysis, and limitations of the prototype that the new study is to fulfill.

2.6.1 Performance

Recent literature has highlighted that response time and integration speed are the most important performance measurements in artificial intelligence systems incorporated into PACS (Picture Archiving and Communication System), especially for emergency as well as high-volume radiology workstations. Joy (2024) said healthcare organizations deploying PACS frameworks with AI algorithms were able to process images and analyze results up to 31% faster than by using a manual approach. For AI-enabled triage systems in emergency situations, detection of critical findings such as pneumothorax was at seconds from the time of image acquisition. The ability to reduce waits of several hours while waiting for radiologist review (having the urgent cases prioritized) and total turnaround time was substantial (Joy, 2024).

Besides the response time, diagnostic consistency was observed to be a key performance gain of AI and PACS integration. Joy (2024) demonstrated that using a standardized AI interpretation model had a 25% reduction in readability variability among the radiologists. Furthermore, diagnostic consistency measurements increased up to 91% for the successfully installed systems and thus repeat readings of the same image were found to provide stable and reproducible results in various clinical settings. This

repeatability is of particular importance in the current practice of radiology where workload and fatigue can lead to non-uniform human interpretation. Joy (2024) pointed to the almost 49% of radiologists with significant burnout, indicating that AI systems can be supportive in offering reliable initial readings. All of these findings in total suggest the ability of AI to make substantial improvements in how fast things are responded to, the consistency of the diagnosis that is arrived at and the efficiency of time spent by the staff in a PACS. It also highlights the importance of having effective technical infrastructure and security aspects to ensure system stability and effectiveness. These findings suggest AI-imbedded PACS systems may provide significant performance benefits in comparison with traditional workflow paths, but their implementation needs to be carefully coordinated to ensure the safety and security of clinical practices.

2.6.2 Result analysis

Analysis of results is compared against earlier PACS generations and current PACS that show a shift from passive storage to active diagnostic support in networked image repositories. The first related work, PACS, by Demner-Fushman et al. (2016) is described as a system with the main purpose of retrieving and distributing radiology examinations, in which images are stored and available for recall by radiologists to facilitate their interpretation but not used computationally during the diagnostic process. In contrast, current AI-facilitated PACS is entering an artificially intelligent workflow where images are being interpreted automatically after they are taken with fully automated interpretation of preliminary measurements or pathologies even before the radiologist opens the study. This results in a “head-start” effect where AI can contribute to the diagnostic pipeline by preparing the case for review rather than being merely a storage system, moving the workflow of routine radiology from a digital filing cabinet to a human-machine partnership Demner-Fushman et al. (2016). However, the gap of clinical acceptance is significant, since

many hospitals are still working with PACS that are not AI-ready because of the technical difficulty of deploying AI solutions in a remote reading PCP using non-standardized DICOM tags differing from one institution to another. These obstacles to integration hinder the capacity for deployment and the operational potency that AI can have in day-to-day usage. Indeed, when AI systems are deployed from the sketch to the simulation of a real hospital environment for clinical use, their operation often suffers because of different data formats and workflow conditions and systems interoperability. This illustrates that, as visionary as their conceptual flow is, current implementations of modern logical flows still need maturity before they can be spread worldwide (Nguyen & others, 2021).

2.6.3 Limitations

Though the recent literature proves the effectiveness of deep learning and Vision-Language Models in the interpretation of the chest X-ray, there are a few limitations to the current research, which are echoed in the presented scope of the study. First, most of the existing literature concentrates on the accuracy of models or the classification, and little emphasis is placed on systems-level integration in clinical processes e.g. within PACS environments. Because of this, it is not clear how many of the proposed models would be practically implemented in a real radiology environment. The workflow simulation is covered in this study, but it is only on a simplified PACS-style setting and without complete hospital integrations like Radiology Information Systems (RIS) and HL7 communication standards. Second, the system generates descriptive and non-diagnostic outputs in order to prevent clinical decision-making. As a result, the resulting reports cannot be truly used in the actual diagnostics and can be inadequate in terms of depth and reliability to be used in the clinical interpretation. This restriction is indicative of the general challenge to current literature on safe usage of generative AI in medical fields. Third, the

proposed system is limited to functional testing, system response , and intergration workflow in the evaluation. There is no assessment of quantitative clinical performance indicators, e.g. sensitivity or specificity. The limitation can be linked to current research which places more emphasis on technical feasibility and explanation rather than direct diagnostic validation. Lastly, the system uses publicly available datasets of chest X-rays and they may not be representative of the diversity and complexity of real-world clinical data. Differences in imaging procedures, patient characteristics, and disease occurrence are thus not exhaustively covered in this research work.

CHAPTER 3

RESEARCH METHODOLOGY

The study uses a system development method to design a prototype used in the interpretation of chest X-rays in a simulated PACS-style process. The system combines a set of pretrained vision-language models and large language models to help analyse images automatically and generate a report. Examples of how the prototype is assessed regarding its feasibility as an AI-assisted diagnostic support system include functionality, performance, and integrating workflow. The methodology will also show how such a system can help improve efficiency and consistency and fit within the normal radiology workflows.

3.1 Methodology Table

Research methodology **table 3.1** containing the objectives of the project, phases, activities, and the deliverables.

Objectives	Phase	Activities	Deliverables
To implement a pretrained Vision-Language Model together with a Large Language Model or generating descriptive, explainable, and clinically meaningful radiology reports from chest X-ray images within a simulated PACS workflow.	Preliminary Study	• Review literature related to chest X-ray diagnosis, Vision-Language Models, Large Language Models and PACS systems. • Analyse the current problems of the related study. • Formulate research problem, objectives, and	Background of Study, Problem statement, research objectives, research questions, research scope, and literature review

		research questions.	
	Data Understanding	- Identify and select a suitable public Chest X-ray dataset from Kaggle website - Review dataset structure and labels	Selected Kaggle Chest X-Ray dataset and dataset description.
	Data Preprocessing	- Inspect images for format consistency - Resize images to ensure compatibility with the prototype system - Apply basic normalization where required for stable inference and visualization	Prepared chest X-ray images ready for inference and system demonstration
	System Design	- Design system architecture (input–process–output) - Define the role of pre-trained BiomedCLIP VLM as a black-box image-to-text inference component - Define the role of the Qwen2.5-1.5B-Instruct, LLM for generating	system architecture diagram, inference workflow design.

		explanatory like output generation • Design inference and output flow within a simulated PACS context	
	System Development	• Configure and utilise pre-trained VLM and LLM via API-based inference (no modification or training) • Design prompt templates for structured, non-diagnostic descriptive outputs • Implement the multimodal inference pipeline	Functional model with AI-generated descriptive text outputs
To develop an integrated PACS-style platform that supports chest X-ray upload, image visualization, AI-assisted disease prediction, report generation, study management, and report export within a single workflow.	System Implementation	• Develop a web-based frontend using Streamlit or Gradio for chest X-ray upload, visualization, and report display. • Implement backend inference services using FastAPI. • Integrate pydicom for medical image handling and OHIF Viewer-style	Streamlit/Gradio-based PACS-style viewer, FastAPI backend services, chest X-ray visualization interface, AI-generated diagnostic report module, MongoDB database.

		image visualization. • Store images and reports using MongoDB.	
To evaluate the complete integrated system in terms of technical functionality, response time, output consistency, system stability, and workflow integration within a simulated PACS-style environment.	Prototype Testing and Evaluation	• Conduct technical functionality testing of the prototype using representative CXR datasets by self-testing. • Measure system performance in terms of consistent output and response time • Evaluate workflow integration	Technical functionality testing results of the proposed prototype, system performance, integration workflow.
	Documentation	• Documenting and Writing the report	Final Year Project report and complete documentation

3.2 Preliminary Study

The preliminary analysis of the system concentrates on laying a solid theoretical and technical basis for the proposed system. Background Review A survey of the literature is performed regarding chest X-ray interpretation, Vision–Language Models (VLM), Large Language Models (LLM), and Picture Archiving and Communication Systems (PACS). This review intends to bring attention to the shortcomings of manual radiograph interpretation, such as diagnostic latency, inter-observer variation, and insufficient number of radiologists. Also, insights from already applied AI-based diagnostic systems are reviewed for recognizing some of the common obstacles regarding explainable solutions and their partial introduction to clinical routine. The review emphasizes that several existing systems are not integrated within the hospital information infrastructure, thus limiting their practical applicability to real settings.

Finally, the literature indicates a research gap in systems combining visual perception and natural language reasoning embedded with a PACS-style workflow. The integration is necessary to allow image organization, visualization, and automatic textual output creation within a common framework. The articles are collated from our arXiv, IEEE eXplore, Pubmed and Scopus. The findings in this stage are used to develop the background of the study, problem statement, objectives of research, questions of research, and scope of research. This phase will be completed with the establishment of a planned path for the later research phases.



Figure 3.1 shows the logo picture of the Arxiv, PubMed, Scopus, IEEE Xplore.

3.3 Data Understanding

The data understanding stage involves the determination of an appropriate public chest X-ray dataset for demonstration and evaluation purposes. The NIH Chest X-ray dataset from Kaggle is utilized in this paper. The dataset includes 112,120 frontal-view chest X-ray images with disease labels. The dataset link is [NIH Chest X-rays](#)

The dataset was chosen because it is a large and public dataset, widely used in the history of chest x-ray study, and suitable for benchmarking and system demonstration with our method. With labeled images, one can also look inside the image content and see which disease categories are being diagnosed.

This part involves the exploration of dataset structure to comprehend image format, labeling methodology, and how the dataset is arranged. The provided sample chest X-rays and corresponding disease labels are reviewed for appropriateness for further preprocessing and system integration. Images and labels of chest X-ray radiographs are shown in Figure 3.1 and in Figure 3.2 respectively. This will result in a curated and well-characterized set of chest x-ray image datasets to be utilized as the main input for model inference and system demonstration towards later stages of this work.



Figure 3.2 shows the example picture of the X-ray in the dataset.

	A	B	C	D	E	F	G	H	I
	Image Index	Finding Labels	Follow-up #	Patient ID	Patient Age	Patient GeView	Posit	OriginalIm	Height
1	00000001_000.png	Cardiomegaly	0	1	58 M	PA		2682	2
2	00000001_001.png	Cardiomegaly Emphysema	1	1	58 M	PA		2894	2
3	00000001_002.png	Cardiomegaly Effusion	2	1	58 M	PA		2500	2
4	00000002_000.png	No Finding	0	2	81 M	PA		2500	2
5	00000003_000.png	Hernia	0	3	81 F	PA		2582	2
6	00000003_001.png	Hernia	1	3	74 F	PA		2500	2
7	00000003_002.png	Hernia	2	3	75 F	PA		2048	2
8	00000003_003.png	Hernia Infiltration	3	3	76 F	PA		2698	2
9	00000003_004.png	Hernia	4	3	77 F	PA		2500	2
10	00000003_005.png	Hernia	5	3	78 F	PA		2686	2
11	00000003_006.png	Hernia	6	3	79 F	PA		2992	2
12	00000003_007.png	Hernia	7	3	80 F	PA		2582	2
13	00000004_000.png	Mass Nodule	0	4	82 M	AP		2500	2

Figure 3.3 shows the example label, which refers to the disease of each picture in the dataset.

3.4 Data Preparation

The data preprocessing stage prepares the chest X-ray images for inference and for system demonstration only. Since the dataset is already collected, preprocessing focuses on image management rather than manual cleaning. All images are first screened for a valid format and resolution, and corrupted, incomplete, or unsupported files are removed. Each image is then resized to the 224 x 224 pixel input resolution required by the Vision-Language Model (BiomedCLIP), so that all images can be processed consistently. Normalization is then applied to bring the pixel values into the standard numerical range expected by BiomedCLIP, using the mean and standard deviation values provided by its `open_clip` preprocessing transform; this keeps the inference stable and reduces the effect of differences in brightness and contrast. Unlike standard deep learning pipelines, no model training or data augmentation is performed in this work, as the system relies only on pretrained models for image-to-text inference. The result of this phase is a set of standardized chest X-ray images that are compatible with the inference pipeline and with PACS-style viewing and system demonstration.

3.5 System Design

The conceptual architecture of the proposed chest X-ray interpretation system is shaped by the system design phase. The system adopts an input–process–output architecture to distinguish the functional roles of the components. Input The input to

the system is chest X-ray images submitted by users via a PACS-style interface. We split the processing stage into two major modules: a Vision–Language Model (VLM) and a Large Language Model (LLM). We instantiate BiomedCLIP as a vision-language model, and it is used as a black-box inference module. It generates visual features from chest X-ray images and preliminary image-conditioned textual representations. We choose BiomedCLIP because of its transformer-based visual encoder and because it has the ability to align image features with language tokens. Since BiomedCLIP generates no fully structured explanatory reports, we propose a Large Language Model for Model-Agnostic Interpretation (Qwen2.5-1.5B-Instruct) to interpret them. The outputs of BiomedCLIP are fed into the Qwen2.5-1.5B-Instruct model using structured prompt templates, which allows for textual reports in a descriptive and interpretable document (findings and impression-style summaries). This modular structure allows the system to go beyond simple image classification in terms of generating human-understandable descriptions based on visual inputs and enables transparent interpretability as the volcanic phenomena are better described.

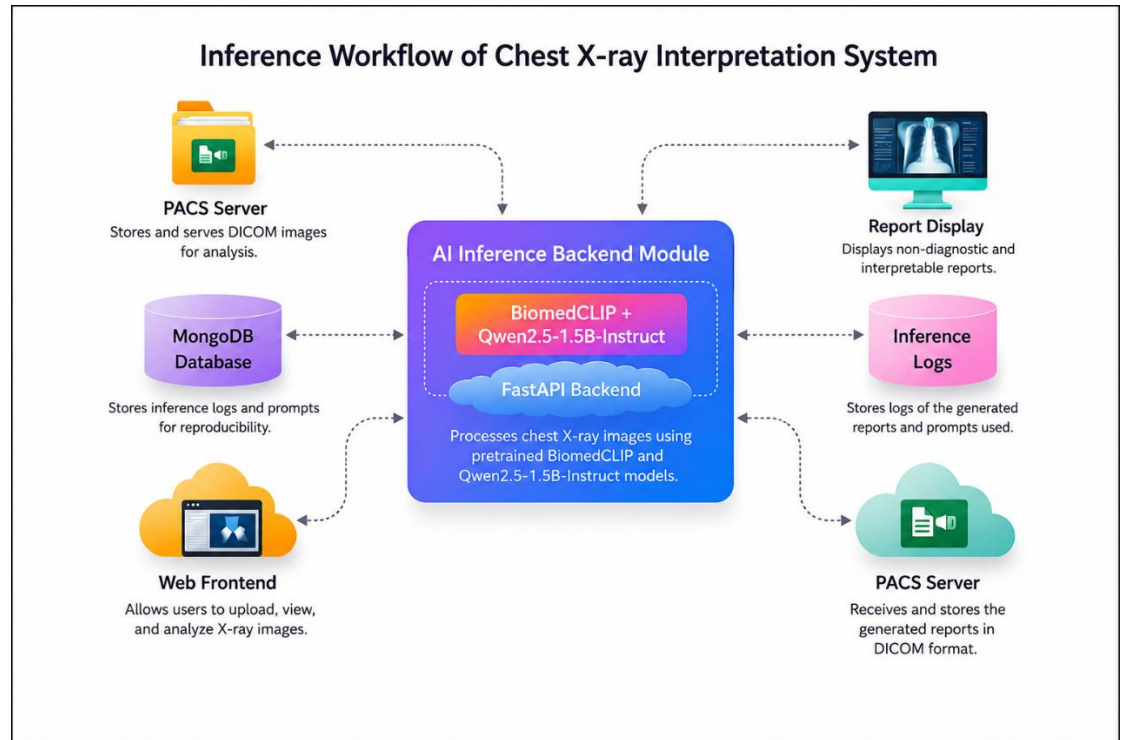


Figure 3.4 shows the inference workflow of the chest X-ray interpretation system.

As depicted in Figure 3.3, the inference workflow begins when the user uploads a chest X-ray image to the web-based PACS-style workstation. The uploaded image is sent to a FastAPI backend service, which is the main processing part of the system. The backend takes the image, processes it accordingly, and passes the processed image to the Vision-Language Model. In BiomedCLIP, the chest X-ray is analysed by comparing its visual features with predefined descriptions of thoracic diseases. This results in disease predictions, confidence scores, and explainability data such as the Grad-CAM image output.

The prediction results obtained from BiomedCLIP are then fed into the prompt template module, which formats the disease finding and confidence information into a structured prompt. The prompt is given to a Large Language Model (LLM) called Qwen2.5-1.5B-Instruct, which generates an easily-readable radiology-style report. Descriptive findings and summary information based on the AI prediction are provided in the report. The MongoDB database can be used to store information related to the inferences, including disease predictions, confidence scores, reports generated in the workflow, prompts, and logs, for retrieval, analysis, and report generated.

Lastly, the report and results are returned to the PACS-style workstation interface, which can be viewed with the uploaded chest X-ray image. The AI generated results can be viewed, a report can be read and downloaded as a PDF. In general, this workflow showcases the seamless integration of the frontend interface, FastAPI backend, BiomedCLIP model, report generation using Qwen model, and the MongoDB database, presenting a comprehensive solution for generating descriptive and interpretable output from chest X-ray images in a simulated PACS (Picture Archiving and Communication System) environment.

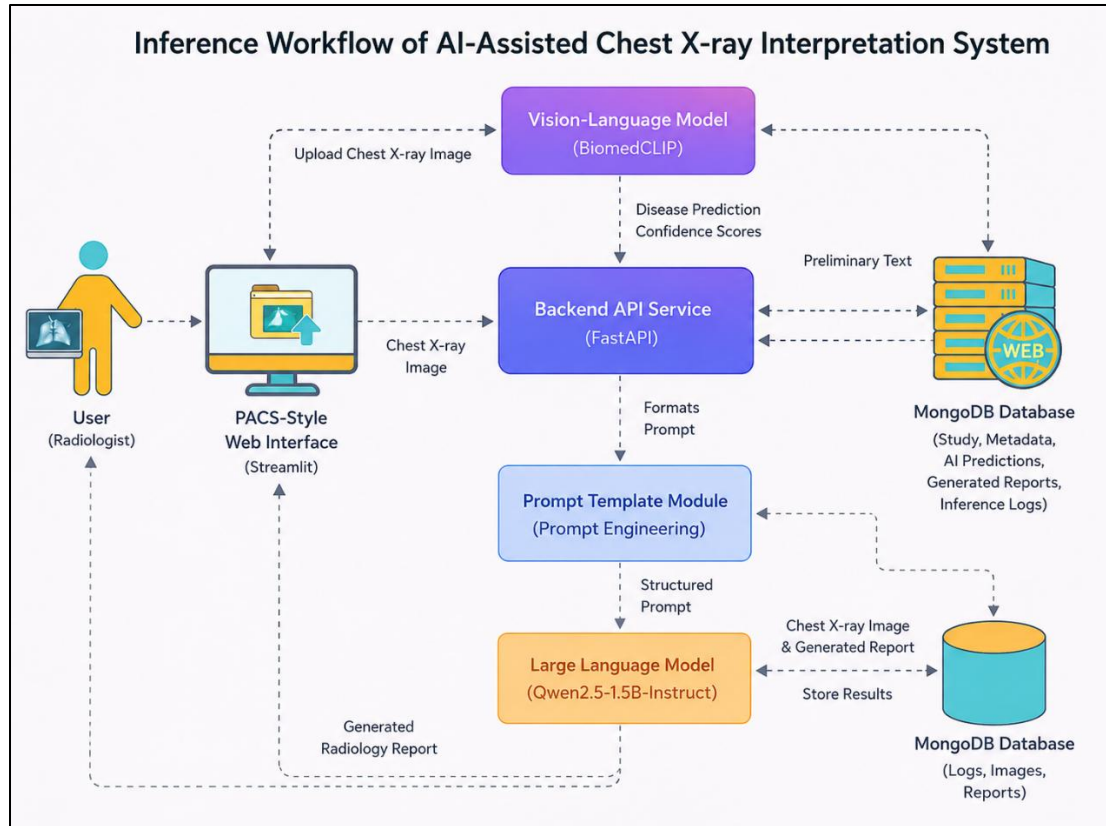


Figure 3.5 shows the system architecture diagram of the integrated PACS prototype.

3.6 System Development

The system development phase moves from the conceptual design to an operational multimodal inference pipeline. Both BiomedCLIP and Qwen2.5-1.5B-Instruct are used as pretrained models without any fine-tuning or retraining, so that the system can be built within the available time and without high-performance computing resources. BiomedCLIP is a Vision-Language Model based on a Vision Transformer (ViT) image encoder together with a text encoder, trained on the PMC-15M biomedical image-text dataset. Rather than generating text itself, BiomedCLIP performs image-text matching: it compares the visual features of a chest X-ray against predefined thoracic disease descriptions and produces disease predictions with confidence scores. To make this matching more robust, several phrasings are defined for each disease and their text features are averaged, a technique known as prompt ensembling, and the most probable

finding together with the top-ranked predictions is selected. In addition, a Grad-CAM heatmap is derived from the model's own similarity score to highlight the image regions that influenced the prediction, providing explainability without adding any extra model. Next, the disease predictions from BiomedCLIP are then fed into a structured prompt template and sent to the Qwen2.5-1.5B-Instruct Large Language Model, which outputs an easily readable radiology report with the sections Technique, Finding, and Impression. In order to keep the output descriptive and non-diagnostic, the prompt is designed that way and a repetition penalty is used in the greedy (deterministic) decoding process, so that the same input will generate the same output. BiomedCLIP and Qwen2.5-1.5B-Instruct are embedded together as a single inference engine, which processes raw chest X-ray images into meaningful textual reports in a fixed sequence: taking in the image, extracting the visual features, building the prompt, and generating a report. This well-defined process keeps the handling of input images and the generation of text outputs stable and reproducible, and supports transparency in the decision-support workflow.

3.7 System Implement

The system phase of implementation includes the integration of a multimodal inference pipeline with a simulated PACS-style workflow. The main purpose of this phase is to show how the proposed AI-driven interpretation system works in clinical practice in a radiology environment. A web-based frontend is built by applying to Streamlit, which include functionalities of uploading chest X-ray images, visualizing the corresponding interpretation, and generating textual outputs produced by AI. Its web-based interface is similar to a basic PACS viewer, providing users with the capability of viewing chest X-rays with generated reports side by side. The backend services are implemented via FastAPI, which handles the reception of images, preprocessing, and running model inferences, as well as sending back the results to the frontend. System modularity, maintainability, and scalability are guaranteed thanks to the fact that frontend and backend components work independently of each other. For standard medical image handling, the pydicom library is used, and it adopts

the visualization layout of the popular OHIF Viewer in clinical applications. MongoDB is used to store the uploaded images and generated textual outputs and is for testing purposes only, and no real patient data are involved. Even without full integrations with hospital systems such as the RIS or competing standards such as HL7, our prototype already well simulates key functions of a PACS uploading. In addition to image upload, visualization, and report generation, the prototype provides supporting modules that are later demonstrated in Chapter 4. Uploaded images are stored in MongoDB using GridFS so that studies remain available between sessions, and a study history (worklist) module allows previously analyzed studies to be searched, filtered, reopened, or deleted, with duplicate detection to avoid storing the same study twice. Generated reports can be exported as PDF documents using the fpdf2 library for documentation and sharing, and a performance dashboard is included to monitor response time and system status. Together, these modules complete the simulated PACS-style workflow implemented in this study.

3.8 Testing and Evaluation

The prototype testing and evaluation stage involves the determination of the technical functionality, system performance, and integration of the proposed system workflow. The major aim of this stage is to ensure that the developed prototype works in a correct and reliable manner as per its design. The representative chest X-ray images of the selected data are used to do technical functionality testing by self-testing to make sure that the whole system is working as planned. This involves ensuring that the system can upload images without crashing, run images through the trained inference pipeline, and produce text-based outputs without crashing. The consistency of the system with repeated image uploads and the inference execution is also monitored to make sure that the system will be capable of running consistently and without crashing. System performance is measured based on the response time taken to produce the outputs, and it is also measured based on the consistency in the textual results generated when used in a series of inferences. The reliability and operational stability of the multimodal inference pipeline are tested with the help of these tests. Workflow integration will be

measured by looking at the functionality of the prototype in providing a PACS-style workflow, which includes image uploading, image visualization, and report presentation within a single interface. The assessment will be made on the system workflow coherence and the technical viability of deploying multimodal AI models to a PACS-style setting. Results of this stage include results of technical functionality tests, results of system performance observation, and results of workflow integration, which will all present the viability and strength of the suggested prototype system.

3.9 Documentation

The documentation phase targeting on combining all methodological explanation discussions into a complete Final Year Project report. This ensures that the research process is transparent, reproducible, and compliant with academic standards.

CHAPTER 4

RESULT AND FINDINGS

The presented results in chapter 4 are results and findings where research results are meaningful by describing the implementation, demonstration, and evaluation of the research prototype.

4 Introduction

The proposed AI-assisted Chest X-ray Diagnosis system result and discussion is discussed in this chapter. The focus is on the implementation, demonstration, and evaluation of the developed prototype, which combines the Vision-Language Model (BiomedCLIP) and the Large Language Model (Qwen2.5-1.5B-Instruct) inside a simulated Picture Archiving and Communication System (PACS) environment. The system aims to support the interpretation of chest X-rays, bringing together an AI component for medical image analysis, an explainable AI component for delivering medical image analysis output, and automated report generation into a single workflow. The primary objective of this chapter is to illustrate the main steps of the proposed system, which process the chest X-ray image, generate the prediction of the diseases, create the structured report with the radiology format, and facilitate the PACS-style workflow. The chapter covers topics like dataset preparation, image preprocessing, system architecture, AI inference pipeline, prototype implementation, performance analysis, and workflow integration evaluation. These findings support the achievement of the research goals set in Chapter 1 and the feasibility of the multimodal integration of AI for interpretation of chest X-rays.

4.1 Chapter Overview

This chapter is subdivided into several sections for the purpose of discussing the developed prototype and its evaluation, in an orderly fashion. The dataset, the preprocessing performed prior to inference and the architecture of the proposed system are explained in Section 4.2. It also includes the entire AI pipeline, including the

collaboration between BiomedCLIP and Qwen2.5-1.5B-Instruct to predict and report diseases. The prototype developed in Section 4.3 is showcased here, using the PACS-style workstation interface, on which image upload, image visualization, AI disease prediction, explainability output, report generation and report export are implemented. Also example cases are demonstrated to represent the performance of the system in various cases. The performance of the system is measured in terms of response time, processing efficiency and output consistency by running the program on various test cases in Section 4.4. Based on the output, the effectiveness and feasibility of the AI-driven process can be assessed. Section 4.5 outlines the technical functionality testing, which is used to test the system for stability, robustness and error handling, to ensure that the prototype functions normally under normal conditions. Finally, Section 4.6 addresses the integration of the proposed AI system in the PACS-style workflow where benefits and limitations, along with practical considerations, are discussed and the appropriateness of the system for supporting radiology tasks is examined. Last, Section 4.7 provides a summary of the key findings from the implementation and evaluation phases and how the prototype meets the goals of this study.

4.2 Dataset and System Overview

This section describes the dataset that has been used to develop the proposed AI-assisted chest X-ray interpretation system, together with the system architecture and preprocessing steps. The purpose is to explain the features of the chosen dataset, the preparation done before inference, and how the whole system operates. Since the prototype only uses pretrained AI models, the preprocessing stage is for preparing images for inference and not for training. The architecture combines the Vision-Language Model (BiomedCLIP), the Large Language Model (Qwen2.5-1.5B-Instruct report generator that runs in-process through Hugging Face transformers), a FastAPI backend service, MongoDB storage with GridFS, and a PACS-style workstation interface built with Streamlit to support the chest X-ray analysis and structured report generation.

4.2.1 Data Description

The dataset used in this project was taken from the NIH ChestX-ray14 dataset, which is one of the largest publicly available chest radiography datasets. It contains more than 100,000 frontal-view chest X-ray images collected from hospital radiology examinations and labelled with fourteen thoracic disease categories, which are Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, and Pneumothorax. Images with no visible abnormality are labelled as “No Finding”, so in total the system recognizes fifteen classes.

Since the full dataset is very large, a representative subset of images was selected for testing, demonstration, and evaluation purposes. The subset includes both normal and abnormal chest radiographs that cover several disease categories. By using a subset, the prototype can still demonstrate its disease prediction and report generation ability while the computational load remains manageable.

The images were obtained in PNG format and were converted into a DICOM-style workflow within the system so that patient metadata such as patient identifier, age, sex, modality, and view position can be carried through the pipeline. The selected dataset is diverse enough to test whether the Vision-Language Model can identify different thoracic abnormalities and produce useful predictions for the report generation stage.

4.2.2 System Architecture Overview

The proposed system applies a modular architecture consisting of five main components, which are the PACS-style frontend workstation, the image preprocessing module, the Vision-Language Model (BiomedCLIP), the

Large Language Model (Qwen2.5-1.5B-Instruct report generator), and the MongoDB database. A FastAPI service acts as the interface between these components, allowing them to communicate with each other.

Workflow starts with the user uploading a chest X-ray image via the web-based PACS workstation. The image is transmitted to the back-end service where it undergoes preprocessing to generate an image suitable for the AI models. The image after the preprocessing is pass to BiomedCLIP Vision-Language Model that performs the comparision of the chest radiograph with the set of thoracic disease descriptions to identify the most probable class of diseases along with the confidence scores. The prediction made by BiomedCLIP is then pass into the report generation module of Qwen2.5-1.5B-Instruct via a well-designed prompt template.

The prediction by BiomedCLIP is then passed through a structured prompt template to the report generation module of Qwen2.5-1.5B-Instruct. The language model generates a comprehensive radiology report based on the detected findings and confidence scores. The processed image stays in GridFS to retain the study between sessions, the report, the disease predictions, the processing metrics and related metadata is stored in MongoDB for later retrieval and analysis.

Lastly, the results are presented in the PACS style workstation interface with the chest X-ray image, confirmed with the AI-generated results, shown with confidence scores, the explainability visualization, and are exported as a PDF file. This architecture offers a full end-to-end workflow that mimics the way that an AI-powered radiology reporting system would operate.

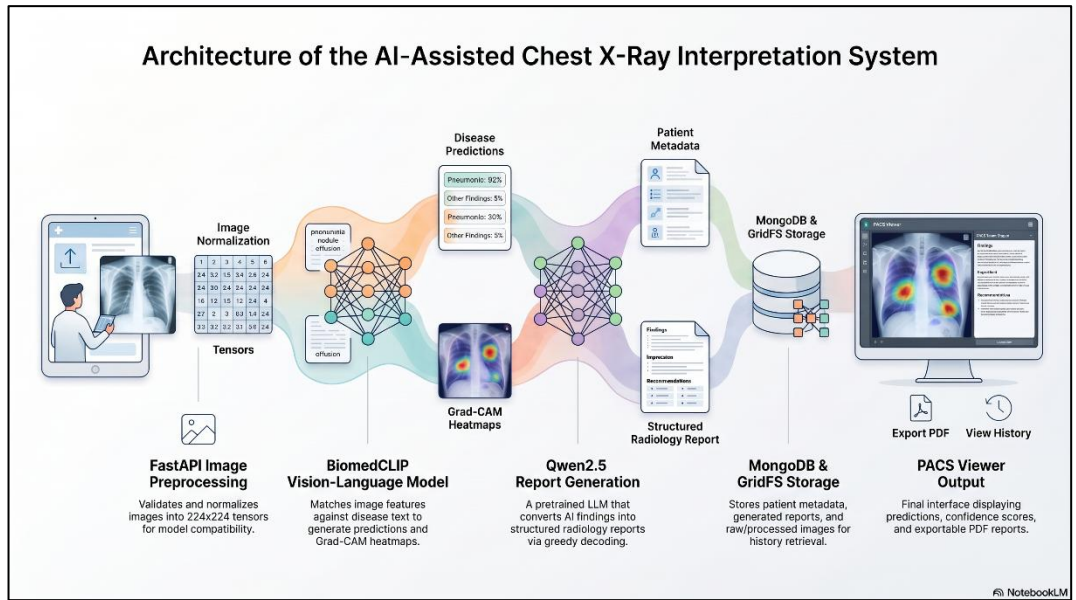


Figure 4.1 shows the proposed system architecture.

4.2.3 Dataset Preprocessing

Chest X-ray images are processed in several steps before AI inference can occur, ensuring consistency in the analysis and compatibility with the AI models. The pre-processing is mainly about preparing and normalizing the images, without any training related transformations, since the system is based on pretrained models.

4.2.3.1 Image Resizing

The original chest X-ray images from the NIH ChestX-ray14 dataset come in different dimensions and resolutions. All images were resized to the standard input size of BiomedCLIP (224 x 224 pixels) prior to making the inference. The size can be standardized, so that the model can process all images in the same way and thus avoid size-related errors. Finally, resizing also helps to improve the computational efficiency as the memory requirement and the

processing overhead are reduced. After resizing, the anatomical structures required for disease prediction and reporting are retained in the resized image.

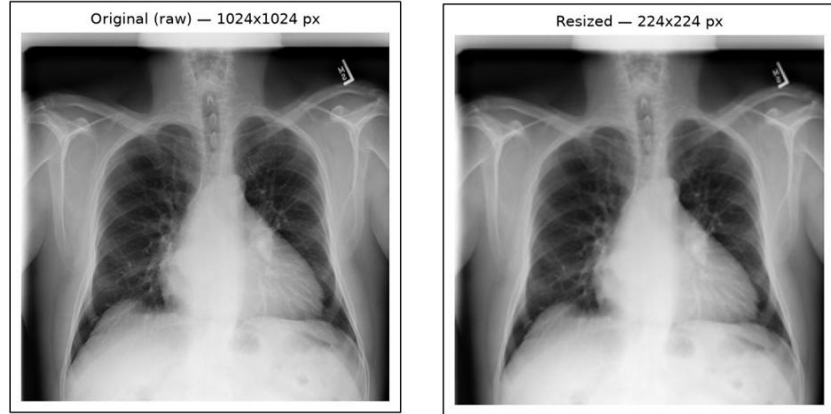


Figure 4.2 shows an example of the image resizing process, comparing the original (raw) radiograph and the resized 224 x 224 image.

4.2.3.2 Normalization Process

The images were resized and then normalized using image normalization to ensure they are compatible with the pretrained BiomedCLIP model. The operation of normalization adjusts the pixel intensity values to a standard range of numbers, based on pre-defined mean and standard deviation values, thus minimizing differences due to image brightness, contrast, and acquisition conditions.

Normalization is an important step for pretrained vision-language models, like BiomedCLIP, which use normalized image inputs for the original training. This consistency in normalization values during the inference process guarantees the stability and reliability of the prediction results of the model, thereby ensuring that the

inference results are as close as possible to the training results. The normalization values in this prototype will be automatically applied by the BiomedCLIP preprocessing transform in the open_clip library.

The normalization process takes the raw pixel values and transforms them into a normalized form before they are fed into the feature extraction layers of the model. This means that the model can be designed on the patterns of images seen in a clinical setting, rather than based upon the difference in image intensity.

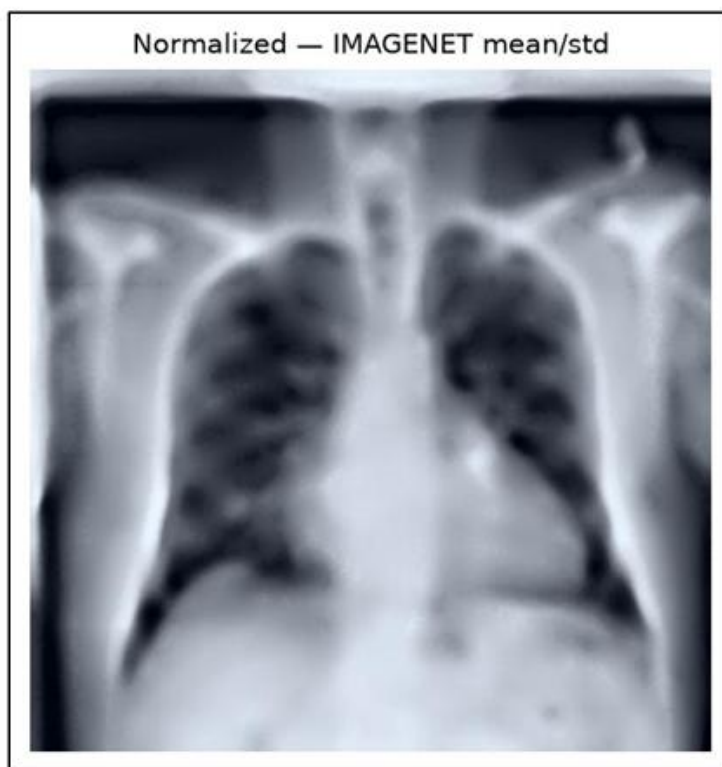


Figure 4.3 shows an illustration of the image normalization process.

4.2.3.3 Data Cleaning and Preparation

Data cleaning was ensured that all the selected images were valid and suitable for data analysis by running a data cleaning process. The evaluation dataset contained corrupted files, incomplete records and unsupported formats, which were eliminated. The images chosen were also reviewed to verify disease labels were consistent with desired testing scenarios. The resulting images were then arranged in a structured data repository for subsequent test and demonstration purposes, ensuring the efficient retrieval of data and smooth running of the overall workflow.

4.2.3.4 Preprocessing Output Observation

After visual inspection of the prepared images, it was observed that all the selected chest radiographs were prepared for AI inference. Important anatomical structures were preserved in the preprocessing steps, which consisted of resizing and normalization, while also ensuring the same size of images and standardized pixel distribution for the entire dataset. The resulting images were sufficiently well visible for disease prediction and explainability analysis. Due to this, the preprocessed dataset provided a stable foundation for the BiomedCLIP-based disease prediction stage and the report generation process using the Qwen model.



Figure 4.4 shows an example of a preprocessed chest X-ray image.

4.2.4 System Pipeline Overview

The proposed system comprises of image preprocessing, disease prediction, explainability analysis, report generation, storing the result in a database, and visualization of the result, which is described in this subsection.

4.2.4.1 Vision-Language Model Processing (BiomedCLIP)

The proposed system includes mainly BiomedCLIP as an image analysis tool. As a Vision-Language Model, it connects the visual features taken from the chest X-ray image with the text descriptions of different thoracic disease categories. The model was chosen because it was trained on PMC-15M, consisting of 15 million biomedical image-text pairs, making it a strong zero-shot matcher for medical images while being a true vision-language model with separate image and text encoders. If a chest X-ray image is uploaded BiomedCLIP automatically detects the image characteristics and

matches them to the predefined descriptions of the diseases. In order to make the matching more robust, several different phrasings are defined for each disease and the text features are also averaged across the different phrasings, a technique referred to as prompt ensembling. The cosine similarity between the image features and every disease representation is then calculated using a softmax function, and the most likely finding is selected and the top 3 predictions are made. The model also generates results of explainability as confidence visualization and Grad-CAM heatmaps besides disease prediction. The Grad-CAM heatmap is generated by computing the gradients of BiomedCLIP's own similarity score with respect to the last vision transformer block, thus identifying the most important parts of the image for the prediction. This technique is a reuse of existing model, no additional model is added to the architecture. Explainability features enhance transparency and enable users to gain insights into the AI model's reasoning.

```
[BiomedCLIP] Predicting diseases...
[BiomedCLIP] Loading : microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224
[BiomedCLIP] Device : cuda
[BiomedCLIP] Trained on PMC-15M biomedical image-text pairs
[BiomedCLIP] Loaded successfully (15 classes)
[BiomedCLIP] Primary disease : Nodule
[BiomedCLIP] Top predictions :
      Nodule           : ##### 81.9%
      Emphysema        : #### 10.4%
      Infiltration      :      2.4%
[Pipeline] Primary Disease : Nodule
```

Figure 4.5 shows the BiomedCLIP disease prediction output, including the primary finding and the top-ranked predictions with confidence scores.

The Grad-CAM visualization provides a visual explanation in addition to the numerical confidence scores, showing the anatomical region that was most influential in the prediction.

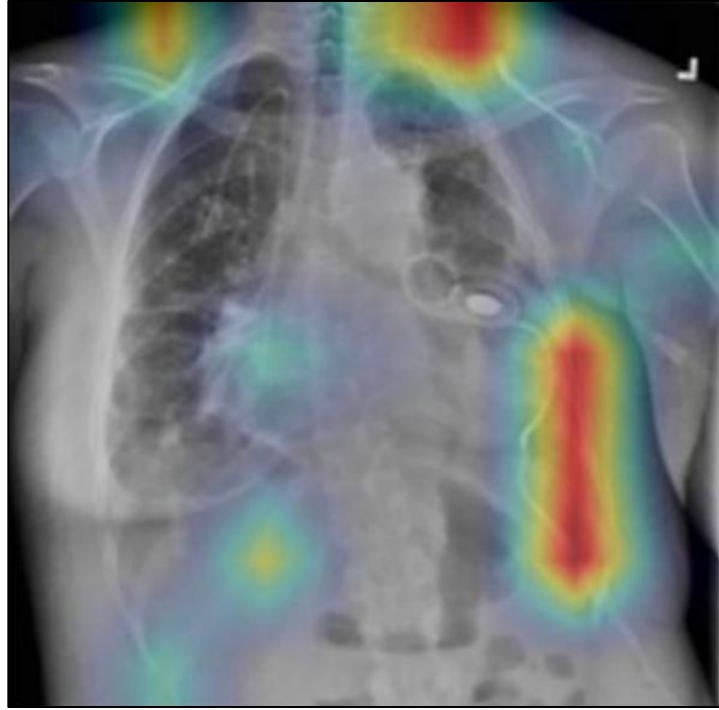


Figure 4.6 shows the Grad-CAM explainability output for the BiomedCLIP prediction.

4.2.4.2 Large Language Model Processing (Qwen2.5-1.5B-Instruct)

Next, the disease prediction is sent to the BiomedCLIP report generation module, which is Qwen2.5-1.5B-Instruct. The intent of this step is to pass the structured prediction result to a clear and easy-to-read radiology style report. The language model is deployed in-process, entirely within the Hugging Face transformers library and loaded as a singleton, thus without the need for an external inference server. The report is generated through a pre-defined prompt template. The prompt contains the predictions for the disease, as well as the high, medium and low confidence levels for each disease prediction, and a brief clinical background for each disease prediction, all extracted from a built-in knowledge base. The model will use this information to create a structured report consisting of 3 parts: Technique, Findings, and Impression. Together with a

repetition penalty, a greedy decoding algorithm produces a deterministic, grounded output, while a post-processing step discards filler phrases or fabricated form fields from the conversation. The report generated from the system is written in a way that makes it appear similar to the way radiology reports are written, but remains firmly based on the BiomedCLIP outputs. This descriptive clinical narrative allows the system to be more easily understood and used by converting the prediction result into a narrative.

```
[Pipeline] Step 2/2 - LLM report generation...
[LLM] Generating report with transformers:Qwen2.5-1.5B-Instruct...
[LLM] Report generated in 15.961s (backend: transformers:Qwen2.5-1.5B-Instruct)
[Pipeline] Step 2 done in 15.961s
```

```
Report preview:
Technique: The examination was performed utilizing standard fronto-lateral views obtained through conventional fluoroscopy without contr
Findings
The imaging study reveals several key features consistent with suspected pneumatosi chyli:
Primary Finding
A significant feature noted within our assessment involves evidence suggestive of "pneumomediastinitis". Specifically observed were signs
Smoke test PASSED - pipeline works on GPU!
```

Figure 4.7 shows the console output of the Qwen2.5-1.5B-Instruct report generation stage.

4.2.4.3 Final Report Generation Process

The last step in the pipeline combines all the outputs into a single report package containing the uploaded chest X-ray image, the disease predictions, the confidence scores, the explainability visualization, and finally the radiology-style report. The final report is presented in the PACS-style workstation-based view for easy viewing of both the visual and textual information in a single view and is also exported in PDF format to document and reference. Study metadata, prediction output, and processing and report generation times are stored in MongoDB to enable data management and retrieval of past cases. The last phase is a single workflow that integrates image analysis, explainability, language generation, and

data storage, showing that multimodal AI can assist with the interpretation and reporting of chest X-ray images.

4.3 Prototype System Demonstration

The developed prototype adopt a PACS style workstation design to deliver one environment for chest X-ray analysis and report generation. The interface was designed to be easy to use for demonstration and evaluation and also to look like a simplified radiology workstation. A number of modules are provided to enhance the image management, disease forecasting, explainability analysis, report generation, performance monitoring, and system configuration.

4.3.1 System Interface Overview

The Home page serves as the main control panel of the system, providing a summary of the system's current status, recent studies, and important metrics. The Worklist module is a study management interface, in which the user can view previous tests and open previously stored reports with search and sorting functions according to their needs. The Upload module enables the user to upload chest X-ray images for analysis, the Viewer module enables image visualization, AI generated findings, AI confidence scores, Grad-CAM explainability output, and AI generated reports. Other modules like Settings page and Performance Dashboard enable system monitoring and setting.

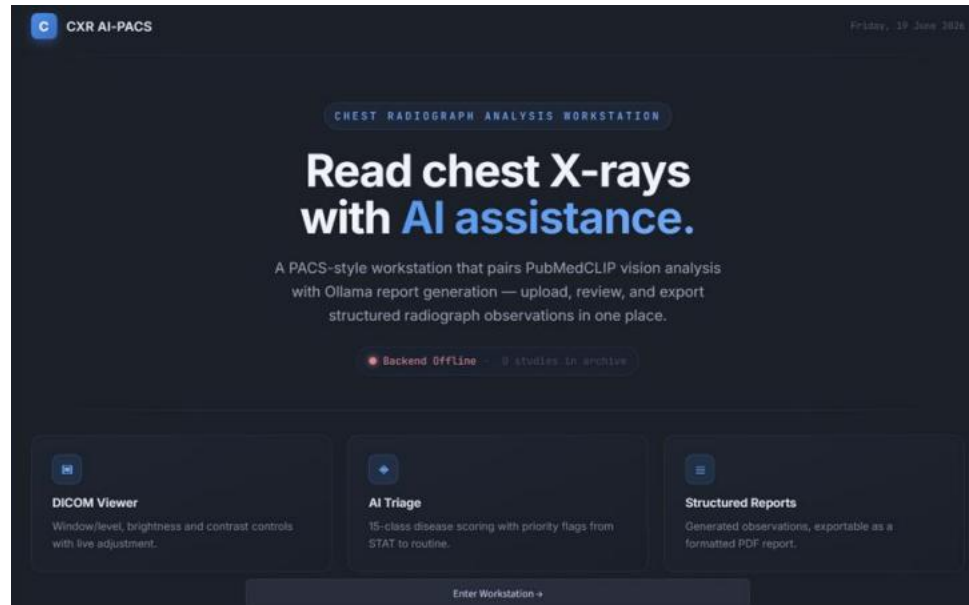


Figure 4.8 shows the home page of the proposed system.

#	FILENAME	STUDY ID	FINDING	PRIORITY	STATUS	DATE	TIME	ACTION
01	00029264_001.dcm	4700405-1	No Finding	NORMAL	ANALYZED	2026-06-19T09:49	15:143z	Open
02	00029064_000.dcm	7670406-1	No Finding	NORMAL	ANALYZED	2026-06-19T09:49	14:87z	Open
03	00028676_029.dcm	1070407-4	No Finding	NORMAL	ANALYZED	2026-06-19T09:48	15:153z	Open
04	00029465_005.dcm	1440408-4	Atelectasis	ROUTINE	ANALYZED	2026-06-18T17:47	15:587z	Open
05	00029264_001.dcm	1440409-8	No Finding	NORMAL	ANALYZED	2026-06-18T17:47	15:576z	Open
06	00029064_000.dcm	1440410-8	No Finding	NORMAL	ANALYZED	2026-06-18T17:46	15:12z	Open
07	00028676_029.dcm	1440411-8	No Finding	NORMAL	ANALYZED	2026-06-18T17:46	14:929z	Open

Figure 4.9 shows the worklist interface displaying available studies, with search and sort functions.

4.3.2 Image Upload and Visualization

The image upload function is where the AI-assisted workflow begins. The user uploads an X-ray image of the chest using the dedicated upload interface, where it is quickly validated before uploaded to the back-end pipeline. Only a list of supported formats (PNG, JPG, JPEG, and DICOM)

and size limits are accepted thanks to the validation. After successful upload, the image is displayed in the viewer where the user can view the radiograph before beginning AI analysis. The viewer helps the user visualize the image, and also creates a clinical setting for reviewing chest X-ray studies, allowing the user to verify the image quality as well as ensure that the right image was selected for inference. Design of the upload workflow aimed at maintaining the simplicity of the user's interaction with the system while still supporting the prototype's PACS-style architecture. The viewer module also provides an variety of image controls to help visualisation. The window width and window level settings allow user to adjust the intensity and contrast of the image, while additional settings include brightness, contrast, image inversion and image saving. The user can visualize the anatomic structures better with these functions and simulate simple viewing tools in PACS workstations.

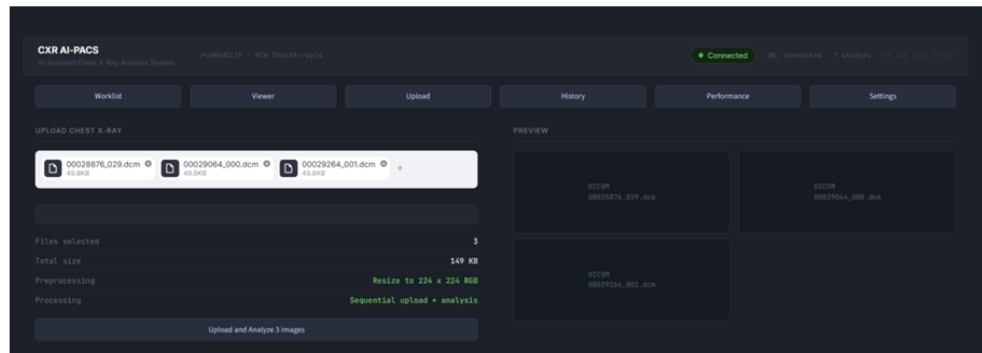


Figure 4.10 shows the chest X-ray image upload interface.

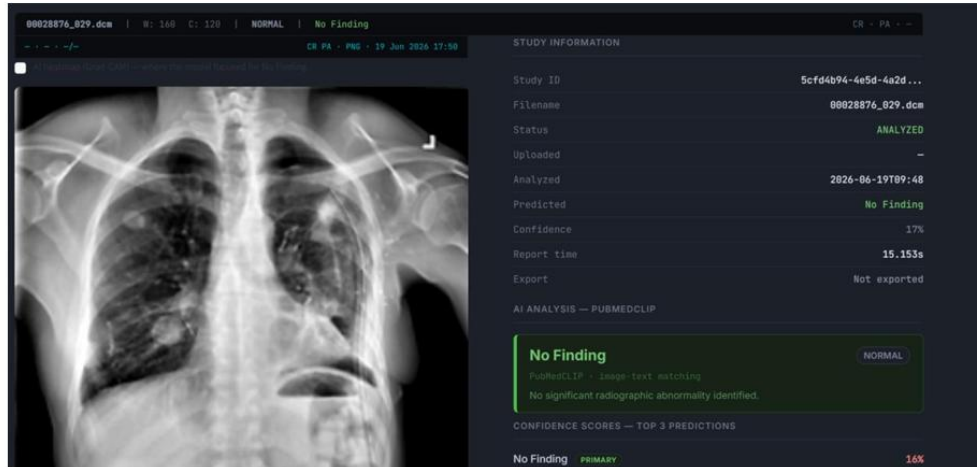


Figure 4.11 shows the chest X-ray visualization within the viewer module.

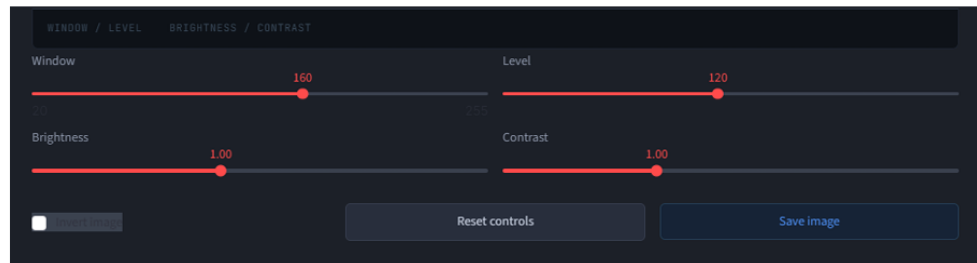


Figure 4.12 shows the image visualization controls, including window/level, brightness, contrast, image inversion, and save image functions.

4.3.3 AI-Generated Output

After the image is uploaded and preprocessed, the system runs the automated disease prediction and report generation using the integrated BiomedCLIP and Qwen2.5-1.5B-Instruct models. The results are displayed in the viewer interface to give the user both visual and textual interpretation of the uploaded chest X-ray. The outputs include the disease predictions, confidence scores, explainability visualization, and a structured radiology-style report, which together give a full overview of the AI analysis and how the system converts image data into meaningful text.

The first stage of the AI analysis is extracting the visual features from the chest X-ray image using the BiomedCLIP Vision-Language Model. Based on these features, the system selects a text description of the most likely finding in the image. The description gives a short summary of the observed radiographic pattern and becomes the basis for the report that follows. By converting the visual information into structured text, the AI-generated findings become easier to interpret and support the later language generation. The descriptions vary depending on the pathology the model identifies and reflect the confidence scores given to the detected abnormalities.

4.3.3.1 AI Findings, Confidence Scores and Interpretation

BiomedCLIP then generates confidence scores for multiple disease categories based on the image's analysis. The disease with the highest score will be considered the main finding and the others will be shown as alternative findings. This rank gives clarity on the decision process of the model and lets the user determine the certainty of each prediction. The viewer interface shows the predicted disease category and its confidence scores. Grad-CAM explainability visualizations are also produced to show the parts of the image that played the most important role in the prediction. These graphical explanations increase readability and make it possible for the user to understand how the model came to its decision. In the case of chest X-ray with Atelectasis, the system correctly classified Atelectasis as the top diagnosis and provided confidence scores for the other diagnoses. The Grad-CAM visualization identified the anatomical areas that contributed to the prediction. In this output you can see that the system can deliver predictive and explainable AI results.

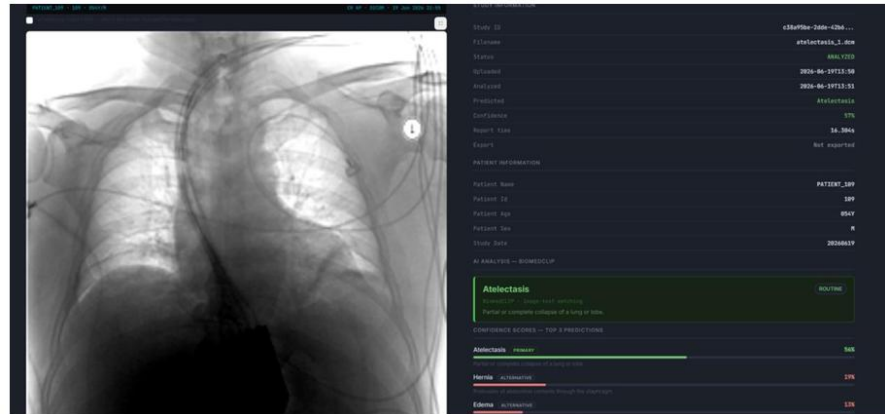


Figure 4.13 shows the disease prediction results and confidence scores generated by BiomedCLIP.

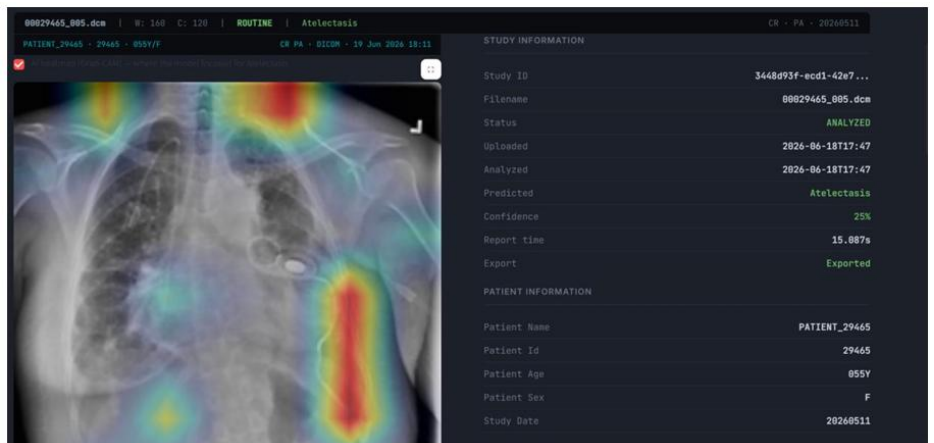


Figure 4.14 shows the Grad-CAM explainability visualization displayed within the viewer.

4.3.4 Report Display and Export

Following disease prediction, the results can be transmitted to the Qwen2.5-1.5B-Instruct report generation module. The report generated is then shown on a viewer interface and can be exported as a PDF for documentation. Report generation transforms the structured prediction output into a clear, radiology-style prose report that can be more easily understood by the user,

extending the application of the system from simple disease classification to providing descriptive text similar to a traditional radiology report.

4.3.4.1 Report Visualization

The report is displayed immediately after analysis in the Workstation interface. It presents itself in a structured format, with several sections: technique, findings and impression. This is useful to ensure clarity and uniformity of output. The report visualization module enables the user to view all of the AI-generated observations as well as the disease predictions and explainability output in a single window for a holistic view and enhanced interpretation of the results. The system also provides a number of report management features to add usability. Using the Edit Report feature, the user can make corrections or additional notes to the generated report before it is exported, and the changes made to the report are recorded in an audit stamp that will be saved with the edited report. The user can also re-run the analysis using the Analysis button or delete records by clicking the Delete Study function. These options offer greater flexibility in reporting and managing AI-generated reports in the PACS-style environment.

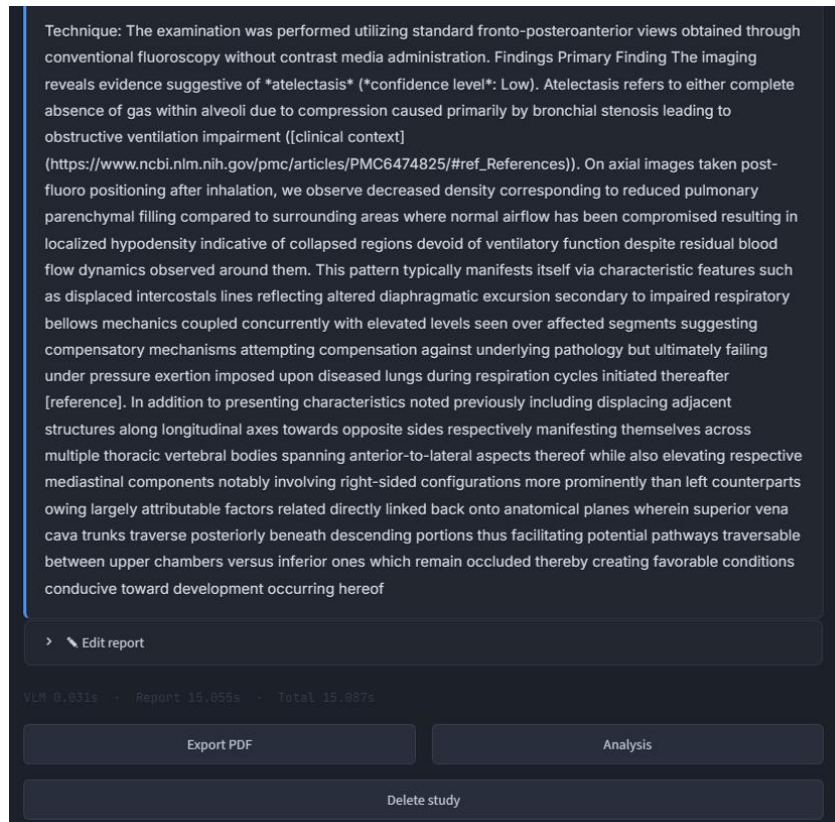


Figure 4.15 shows the AI-generated radiology-style report displayed within the viewer.

4.3.4.2 PDF Report Export Function

The system provides the function to export to PDF with the library `fpdf2` to facilitate documentation and sharing of results. Once the analysis is finished, the user can download the PDF report with the uploaded image, the disease predictions, the confidence scores, the chest X-ray with the heatmap, the radiology style report generated by the system, the system performance metrics and a disclaimer stating that the output is for demonstration purposes only. The exported PDF is a portable copy of the analysis performed and enables the user to save and distribute the result without requiring

the web app, enhancing the usefulness of the system and facilitating post-fact retrieval of results.

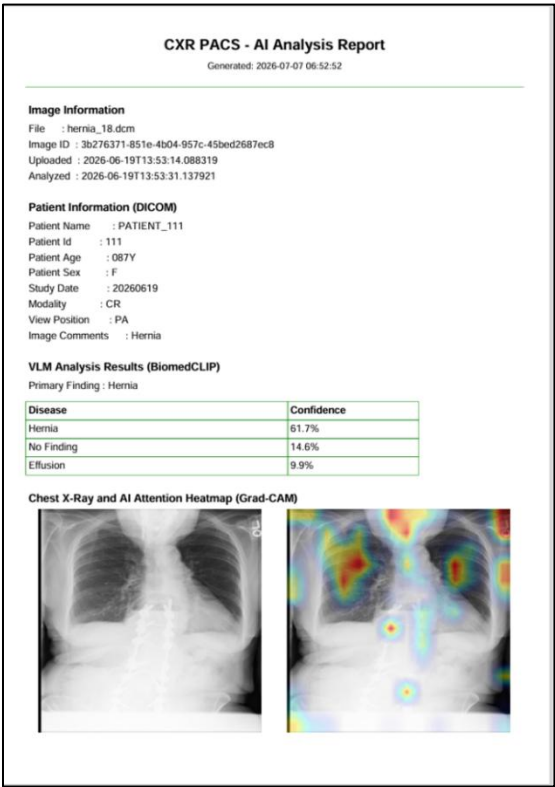


Figure 4.16 shows an example 2 page of the exported PDF report.

4.3.5 Study History Management

The Study History Management module has been created to store and manage previously analyzed chest X-ray studies. It can be used to view past records without re-uploading and reanalyzing the same image. The information saved in each study includes the name of the file, the predicted finding, the priority level and analysis status, allowing the user to see the results of previous studies effectively. Search and filter functions are provided on the interface to permit specific search by filename or disease finding. This will enhance the organisation of data and facilitate the retrieval of relevant cases from the database, particularly after many studies are added to the database. The module also gives the user the ability to delete the selected records, with a Delete Selected function, which can be used to delete unwanted, duplicated and outdated records. Another feature of this is a duplicate-detection mechanism to ensure that the same study is not stored more than once, keeping the database organized and only storing relevant studies. The History module enhances the PACS-style workflow by facilitating retrieval of studies, storage and management of results created with AI, and long-term storage.

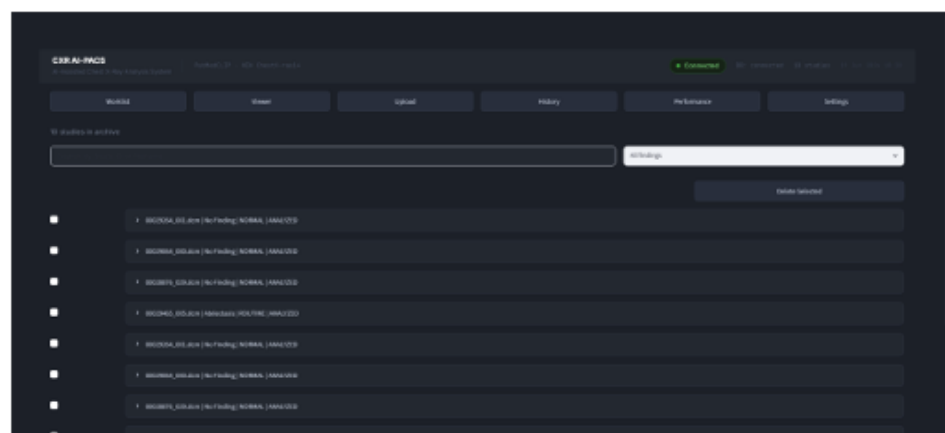


Figure 4.17 shows the Study History Management interface displaying stored studies, search and filtering functions, and record deletion capability.

4.4 System Performance Analysis

This part assesses the effectiveness of the proposed system for AI-assisted chest X-ray interpretation. The evaluation is based on the time required to achieve the inference workflow and consistency of the output generated. Analysis is important because it will indicate if the system can provide timely and reliable results and maintain stability while going through the analysis. The studies that were evaluated included 10 chest X-rays from various categories of thoracic diseases: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Pleural Thickening, and Pneumothorax.

4.4.1 Response Time Evaluation and Result

The time that the system takes to process a chest X-ray picture and produce the final report. Evaluation carried out to measure the efficiency of the workflow and to find out which components take the maximum time in the workflow. The response time analysis was conducted on ten chest X-ray studies that went through the entire AI inference pipeline, including disease prediction, report generation, storing the results in a database, and presenting the final result. The table 4.1 below includes LLM processing time and database processing except for the VLM because the VLM for all tests is 0 (s).

Table 4.1: Processing time results for ten chest X-ray studies.

Study	Predicted Finding	LLM (s)	DB (s)	Total (s)
1	Pleural Thickening	16.0	0.02	16.0
2	Pneumothorax	16.7	0.01	16.7
3	Atelectasis	16.3	0.01	16.3
4	Cardiomegaly	15.8	0.01	15.8
5	Consolidation	15.9	0.01	15.9
6	Edema	15.7	0.01	15.7
7	Effusion	15.8	0.01	15.8
8	Emphysema	16.5	0.01	16.5
9	Fibrosis	16.0	0.01	16.0
10	Hernia	15.8	0.01	15.8

Table 4.2: Summary of system performance.

Metric	Value
Number of Studies	10
Average Processing Time	16.05 s
Fastest Processing Time	15.7 s
Slowest Processing Time	16.7 s
Average Database Time	0.011 s
Processing Time Range	1.0 s

The Processing Time noted for each study is shown in Figure 4.18. The results show that the system has relatively stable performance in all cases. The average total processing time was about 16.05 seconds per study, the fastest study being analysed 15.7 seconds, and the slowest study 16.7 seconds. There was only about 1 second difference between the fastest and slowest runs from one study to the next. This implies that processing load remains consistent across different disease categories to be analyzed, and the performance of the system remains predictable with various chest X-ray abnormalities. The results also demonstrate that the most time consuming portion of the overall processing time was spent generating reports. In all studies the LLM processing time ranged from 15.7 to 16.7 seconds and the database operations took between 0.01 and 0.02 seconds, a negligible time. The generation of reports is thus the predominant computational concern of the current implementation.

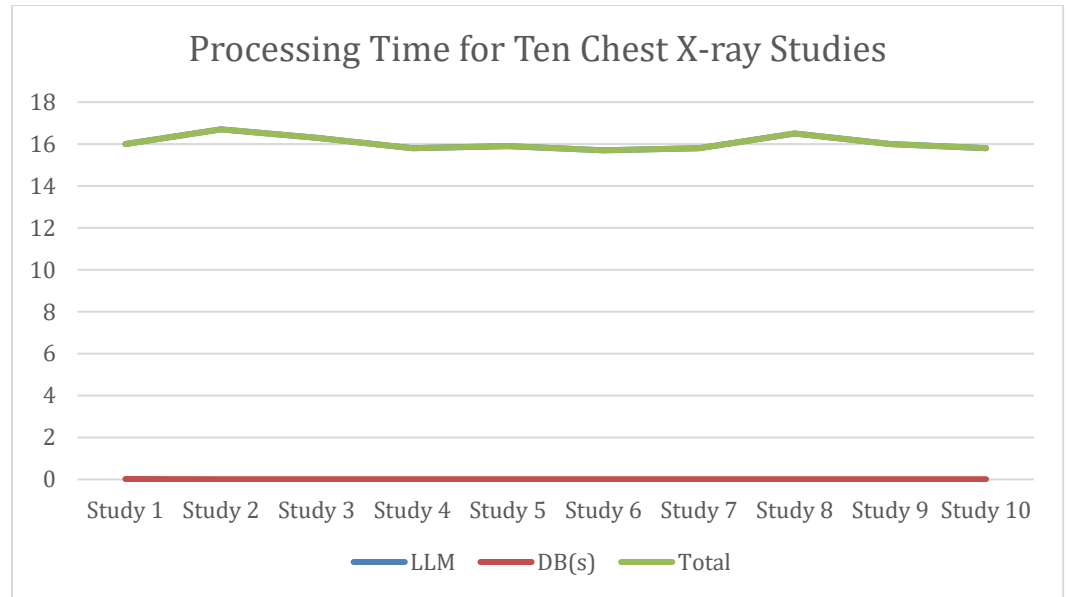


Figure 4.18 shows the line chart of per-study processing times recorded during system evaluation.

4.4.2 Discussion of Performance

The performance evaluation shows the system can produce AI-assisted chest X-ray reports within a consistent timeframe. The full workflow takes about sixteen seconds, which is acceptable for research and demonstration purposes, especially given how complex the report generation is. Compared with earlier work, the measured average response time of 16.05 seconds lines up with the efficiency gains reported in the literature. Joy (2024) reported about 31% faster image processing when adding AI inference to a PACS-style workflow. The stable per-study times seen here (15.7-16.7 s) also show that placing AI inference directly in the workflow does not add excessive delay, and this study goes a step further by generating full natural-language reports. One important observation is that the Vision-Language Model adds very little processing overhead compared with the Large Language Model. Disease prediction finishes almost instantly, while report generation needs much more computation because it produces detailed text. Because of this, most of the total response time comes from natural

language generation, not image analysis. The database layer was very effective during the evaluation. The storage time is always below 0.02 seconds in all studies, indicating that the MongoDB integration has no noticeable impact on system responsiveness. This implies that the study metadata, prediction results, and generated reports are stored efficiently in the database. The overall response time evaluation demonstrates a consistent and reliable response time for the system, across all chest X-ray examinations performed. The relatively small difference in processing times indicates steady processing behavior, and demonstrates that the workflow is suitable for AI-assisted radiology reporting in a PACS-like setup.

4.4.3 Output Consistency and Reliability

The output consistency evaluation checked whether the system gives stable and reproducible results when the same image is analysed more than once. Consistency matters for an AI system because reliable output builds user confidence and shows the pipeline behaves predictably under the same conditions. To test this, the same chest X-ray image containing Atelectasis was analysed five times, and the disease predictions, confidence scores, and processing times were recorded for each run. The goal was to see whether repeated analysis gave the same findings and similar performance.

Table 4.3: Repeated testing results using the same chest X-ray image.

Test Iteration	Predicted Finding	Confidence Score	Total Processing Time (s)
Test 1	Atelectasis	56.9%	16.23
Test 2	Atelectasis	56.9%	15.68
Test 3	Atelectasis	56.9%	15.44
Test 4	Atelectasis	56.9%	16.30
Test 5	Atelectasis	56.9%	15.94

As shown in the results in Table 4.3, the system correctly identified Atelectasis as the major finding in all five runs of the experiment. The confidence score remained constant at 56.9%, and the order of the other predictions did not change either. No change was seen in the disease classification, which shows stable inference behaviour for identical input. Processing times varied slightly, but not enough to affect the prediction. The times recorded were from 15.44 seconds to 16.30 seconds, which is a difference of less than 1 second between all runs.

4.4.4 Consistency Analysis

The consistency evaluation shows the system can produce consistent output for identical chest X-ray images. In all five runs, the Vision-Language Model has always been able to accurately detect the disease Atelectasis and always provided the same confidence scores, indicating that the disease prediction task has a very stable performance and does not introduce a lot of randomness in the inference process. The reports generated were consistent across all runs, beyond the disease predictions. The small variations in processing time were due to system resource allocation and execution conditions and did not affect the final prediction or report content, thereby not changing the overall interpretation. The uniformity of the two stages, disease prediction and disease report generation, reflects the stability of the AI-assisted workflow. The outcomes indicate that the output of the hybrid pipeline (BiomedCLIP+Qwen) is reliable when the input images are

the same. In medical imaging, consistency is a key factor, as reproducible results are essential for reliable and trustworthy behaviour. A hand-coded review of the observational reports was also conducted for all five runs. The clinical significance of the report and the significance of the disease remained unchanged. Despite the use of a Large Language Model in the report module, there was no significant difference in the report generated for the same chest X-ray image. This can be attributed to greedy (determinative) decoding and a limited prompt, which is highly indicative of the effectiveness of the prompt engineering method in generating consistent and repeatable reports. The prototype is highly consistent and reliable overall with repeated testing. The absence of prediction variations over multiple runs is indicative of the stability and research and demonstration readiness of the inference pipeline.

4.5 Technical Functionality Testing

This section presents the technical functionality testing done to check the stability, robustness, and error handling of the developed prototype. The purpose is to confirm the system works correctly under normal use and responds gracefully when invalid input or unexpected conditions occur.

4.5.1 Functional Testing Procedure

A dedicated testing endpoint was used in the FastAPI backend for functional testing. The endpoint executes a sequence of tests to test the key components of the system: sample image availability, image loading, BiomedCLIP inference, validation of the output structure, Qwen report generation, end-to-end pipeline execution, API health, and database connectivity. A PASS, FAIL, or SKIP result is obtained for each test and a short diagnostic message is provided, and a summary indicating the number of tests that passed is presented. All components were returned as PASSED

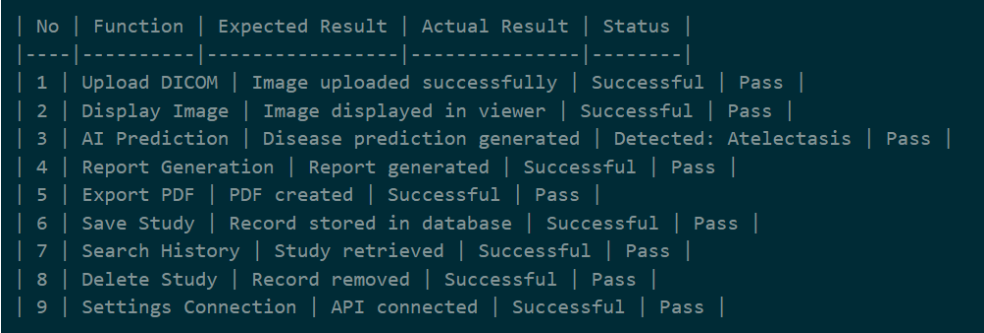
during testing. The image succeeded in loading, BiomedCLIP returned a valid primary finding with a confidence score ranging from 0 to 1, and the Qwen module returned a non-empty structured report, while the overall pipeline did not have any errors. The health checks against the API and database also verified that the service remained responsive and records were saved properly. The data demonstrates the individual modules and the pipeline as a whole working as expected.

Table 4.4: Functional testing results.

No	Function	Expected Result	Actual Result	Status
1	Upload DICOM	Image uploaded successfully	Successful	Pass
2	Display Image	Image displayed in viewer	Successful	Pass
3	AI Prediction	Disease prediction generated	Successful	Pass
4	Report Generation	Report generated	Successful	Pass
5	Export PDF	PDF created	Successful	Pass
6	Save Study	Record stored in database	Successful	Pass
7	Search History	Study retrieved	Successful	Pass
8	Delete Study	Record removed	Successful	Pass
9	Settings Connection	API connected	Successful	Pass

All major components functioned as required by functional testing. Each feature was tested individually and compared with the expected results. As indicated in Table 4.4, it can be seen that there were no functional failures for all the functions that have been tested, meaning that the system can

successfully upload DICOM, Display Image, AI Prediction, report generation, export PDF, save study, search history, delete study and settings connection. There were no failures noted in all nine functional test cases.



No	Function	Expected Result	Actual Result	Status
1	Upload DICOM	Image uploaded successfully	Successful	Pass
2	Display Image	Image displayed in viewer	Successful	Pass
3	AI Prediction	Disease prediction generated	Detected: Atelectasis	Pass
4	Report Generation	Report generated	Successful	Pass
5	Export PDF	PDF created	Successful	Pass
6	Save Study	Record stored in database	Successful	Pass
7	Search History	Study retrieved	Successful	Pass
8	Delete Study	Record removed	Successful	Pass
9	Settings Connection	API connected	Successful	Pass

Figure 4.19 shows the functional testing results returned by the system test endpoint.

4.5.2 System Stability

The stability test determined if the system would crash when processing several requests. The prototype uploaded and analysed a series of chest x-ray studies, sequentially, during the test. The backend tracks the total number of requests, the number of errors, the error rate, and system uptime, which together show the overall stability. The system did not experience any unexpected shutdown during the test and the system continued to run throughout each study. The models remained in memory as singletons after the initial load, which meant later analyses did not add to the loading overhead and the service remained responsive. This system was tested by uploading 10 X-ray images and observing if the prototype would crash or not.

```

- DICOM images processed : **10**
- Successful analyses    : **10**
- Crashes                : **0**
- System interruptions   : **0**

```

Figure 4.20 shows the result after evaluating 10 images and analyses at once.

4.5.2.1 Error Handling Evaluation

Error handling was tested by deliberately feeding the system invalid input. When an unsupported file type was uploaded, the backend rejected the request and returned a clear error message instead of crashing. Oversized files were rejected based on the configured file size limit, and DICOM files that could not be decoded produced an informative error instead of an unviewable output. A request for a non-existent image returned a controlled not-found response.

The pipeline itself was also designed to fail safely. If BiomedCLIP inference does not succeed, the pipeline raises a controlled error instead of generating a misleading report, and if the language model returns unusable output, the system reports the failure instead of showing an empty report. These mechanisms keep errors contained and clearly communicated to the user.

Table 4.5: Error handling results.

Scenario	Expected Behaviour	Result
Unsupported file	Rejected	Pass
API offline	Warning displayed	Pass
No file selected	Upload prevented	Pass

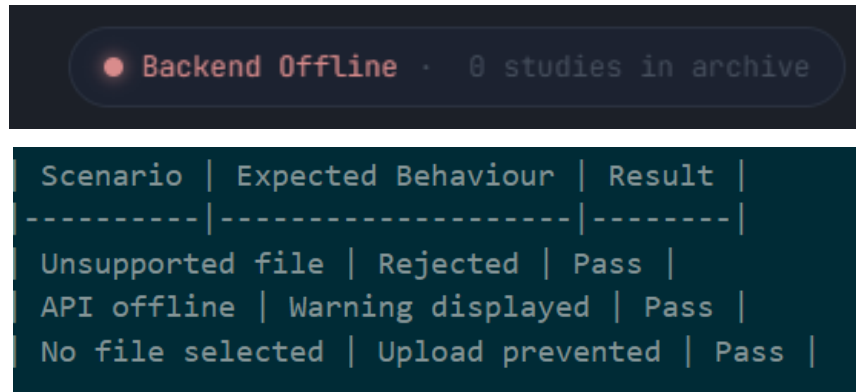


Figure 4.21 shows an example of graceful error handling for an invalid upload.

4.5.2.2 System Stability Analysis

The stability test demonstrated that the system remained stable during repeated testing sessions. There were no crashes, database errors, or report generation interruptions during all studies uploaded, and error handling was appropriately constructed to deal with invalid inputs and connection issues. The results suggest that the prototype is sufficiently stable for research and demonstration. As indicated in Table 4.5, all errors handling scenarios were completed with a Pass status, thus indicating all functional and stability testing were passed.

4.6 Workflow Integration Evaluation

This is about what is being proposed by the AI system and how it integrates into the PACS workflow. The evaluation process includes a workflow evaluation, its integration with the PACS-style interface, and the benefits and limitation from the integration.

4.6.1 End-to-End Workflow Validation

The proposed workflow follows a logical sequence that mirrors a simplified radiology reporting process. A chest X-ray image is first uploaded through the workstation and preprocessed by the backend. BiomedCLIP then analyses it to get the disease predictions and confidence scores, after which a Grad-CAM heatmap is generated to explain the prediction. The findings are passed to the Qwen2.5-1.5B-Instruct model, which produces the structured observational report. The image, predictions, explainability output, and report are then stored in the database and shown to the user, who can review, edit, or export the report. This end-to-end flow combines image analysis, explainability, language generation, and data storage into one coherent workflow.

Upload Preprocessing → BiomedCLIP (Vision-Language Model) → Report Generation (Qwen2.5-1.5B-Instruct) → Database (MongoDB) → History → PDF Export
--

To check the end-to-end functionality, a chest X-ray study was processed through the complete pipeline shown above. Each stage ran in sequence and passed its output to the next module without manual intervention, confirming the whole workflow works as a single integrated system.

4.6.2 Integration with PACS-style Workflow

The proposed AI pipeline was integrated into a PACS-style workstation interface made up of several modules, including the Home dashboard, Worklist, Upload, Viewer, Performance Dashboard, History, and Settings. This modular layout mirrors a conventional PACS workstation and lets the AI-generated output be reviewed next to the chest X-ray image in a familiar layout. The integration shows that modern vision-language and language

models can be built into a radiology-style workflow without breaking the usual sequence of image review, interpretation, and reporting.

Table 4.6: Module integration results across the workflow.

Module	Integration Result
Upload	Successfully sent image to backend
BiomedCLIP	Generated prediction
LLM (Qwen2.5-1.5B-Instruct)	Generated report
MongoDB	Stored study
History	Retrieved study
PDF	Successfully exported

4.6.3 Discussion of Workflow Integration

The workflow integration evaluation produced several observations taken directly from testing the complete system. All modules communicated successfully, with each stage passing its output correctly to the next, and no failures occurred during the complete workflow execution. This shows the integration between the upload interface, inference models, database, and export functions is stable. The workflow also combined image analysis, report generation, database storage, and report export into one coherent process. As shown in the performance analysis, the overall processing time was driven mainly by the LLM (report generation) stage rather than image upload, inference, or database operations. Based on these points, integrating multimodal AI into a PACS-style environment is feasible.

A few limitations should also be noted. The evaluation used a representative subset of the dataset rather than the full ChestX-ray14 collection, and the models are used in a zero-shot, inference-only way without fine-tuning on the target dataset. The report generation stage, which uses a 1.5-billion-parameter language model, accounts for most of the roughly sixteen-second response time and focuses on a single primary finding. Most importantly, the prototype is intended for research and demonstration only and is not validated for clinical use, so all AI-generated output needs to be confirmed by a qualified radiologist

4.7 Summary

This chapter presented the results and discussion of the proposed AI-assisted chest X-ray diagnosis system. It described the dataset and preprocessing steps, the system architecture and inference pipeline, the prototype demonstration, the performance analysis, the technical functionality testing, and the workflow integration evaluation. Based on the results, combining BiomedCLIP and Qwen2.5-1.5B-Instruct in a PACS-style environment is feasible and produces consistent, explainable, and exportable radiology-style reports.

4.7.1 Key Findings

The key findings of this chapter can be summarised as follows. The system successfully processes chest X-ray images and produces disease predictions with confidence scores using the BiomedCLIP Vision-Language Model, supported by Grad-CAM explainability. The Qwen2.5-1.5B-Instruct model then takes these predictions, structures them into radiology-style reports and documents is viewable and editable as a pdf file. This system maintains stable and predictable performance, with an average processing time of around 16.05 seconds per study, and no significant overuse in the database, and repeated tests indicated that the system generates repeatable predictions

and reports for same inputs. The functional and stability test also showed a successful performance of the prototype; error and functional test cases was run successfully with all test cases being returned as Pass. Lastly, the end-to-end workflow validation demonstrated that each module was communicating successfully and that the complete pipeline executed successfully without failure, thereby all the steps of the image analysis, report generation, database storage and report export were well integrated into a PACS-like environment. The results validate the research aims and present the feasibility of incorporating multimodal AI in the interpretation of chest X-rays.

CHAPTER 5

CONCLUSION LIMITATION & FUTURE WORK

This chapter gives the general result of the study, its limitation on the proposed system and recommendations for future work. It summarizes the research as per the objectives in Chapter 1 and the implementation and evaluation outcomes in Chapter 4.

5.1 Introduction

This chapter also highlights the main achievements and contributions of the developed AI-assisted chest X-ray interpretation system within a simulated Picture Archiving and Communication System (PACS) workflow.

The conclusion in this chapter looks at how well the research objectives were achieved and discusses the significance of the developed prototype. The chapter also contains implementation and evaluation constraints that provide the reader with the real-world capabilities and deficiencies of the prototype. Lastly, several recommendations are provided for future work to enhance the model performance of the system, add functionalities, and facilitate research in medical image analysis and radiology reporting using AI. The chapter summarizes the most important results of the study and provides guidance for future evolution of AI-based diagnostic systems.

5.2 Conclusion

This study was carried out to develop and evaluate an AI-assisted chest X-ray interpretation system, combining a Vision-Language Model (BiomedCLIP) and a Large Language Model (Qwen2.5-1.5B-Instruct) within a simulated Picture Archiving and Communication System (PACS) workflow. The developed prototype combined chest X-ray visualisation, AI-assisted disease prediction, explainable visual output,

automated report generation, study management, and PDF export in one web-based platform.

The system was evaluated and found to be capable of performing the desired workflow. The prototype demonstrated stable performance during functionality testing, consistent output with the same chest X-ray images analysed multiple times and the AI inference pipeline was integrated into a PACS-like environment. The findings showed that the system is a viable proof-of-concept for AI-assisted chest X-ray interpretation. Overall, this research shows that combining a Vision-Language Model and a Large Language Model into an intelligent medical imaging workflow is feasible. Although the system is mainly intended for research and educational use, it lays a foundation for future work on AI-assisted radiology reporting and intelligent medical imaging systems.

5.2.1 Achievement of Research Objectives

The first research objective was to implement a pretrained Vision-Language Model (BiomedCLIP) together with a Large Language Model (Qwen2.5-1.5B-Instruct) to generate descriptive, explainable, and clinically meaningful radiology reports from chest X-ray images within a simulated PACS workflow. This objective was achieved. BiomedCLIP analysed the uploaded chest X-ray images to produce disease predictions with confidence scores and Grad-CAM explainability visualisations, and Qwen2.5-1.5B-Instruct converted these predictions into structured radiology-style reports. Together they show that medical image analysis and explainable report generation can be integrated into one workflow.

The second research objective was to develop an integrated PACS-style platform that supports chest X-ray upload, image visualisation, AI-assisted disease prediction, report generation, study management, and report export within one workflow. This objective was met through a web-based

workstation made up of several integrated modules: Home, Worklist, Upload, Viewer, History, Performance, and Settings. The system lets the user upload chest X-ray images, view the images, generate AI-assisted reports, manage past studies, and export structured reports as PDF files, which shows a unified PACS-style workflow.

The third research objective was to evaluate the complete integrated system in terms of technical functionality, response time, output consistency, system stability, and workflow integration within a simulated PACS-style environment. This objective was met through a thorough evaluation covering functionality testing, response time analysis, output consistency testing, system stability evaluation, and workflow integration assessment. The results showed the prototype ran reliably, kept consistent output during repeated testing, and successfully integrated all components into a complete AI-assisted chest X-ray interpretation workflow.

5.2.2 Research Contributions

This study contributes to the development of AI-assisted medical imaging systems by showing that a Vision-Language Model and a Large Language Model can be integrated within a simulated PACS-style environment. Unlike a conventional image classification prototype that only focuses on disease prediction, the proposed system combines chest X-ray visualisation, explainable AI, automated report generation, database management, and PDF report export in one workflow. This gives a fuller demonstration of how AI can support chest X-ray interpretation in a practical medical imaging setting.

The study also shows that pre-trained foundation models can be used for AI-assisted chest X-ray interpretation without extra model training or fine-tuning. Using BiomedCLIP for disease prediction and Qwen2.5-1.5B-

Instruct for report generation shows that modern foundation models can be built into intelligent healthcare applications through inference-based deployment alone. The developed prototype also lays a foundation for future research on explainable AI, intelligent radiology reporting, and AI-assisted clinical decision-support systems.

5.2.3 Lessons Learned

Several important lessons came out of developing this system. Integrating two independently pre-trained foundation models, BiomedCLIP for vision and Qwen2.5-1.5B-Instruct for language, required careful handling of how they talk to each other. The quality of the generated report depends directly on the disease label passed from the vision stage, so a weakness in one model carries over to the other. The project also showed how important deterministic decoding is in medical AI. Greedy decoding was needed not just to avoid numerical instabilities during generation, but also to guarantee reproducible reports, which is a basic requirement for clinical trust. These lessons show that in medical applications, engineering choices around reproducibility and how models connect to each other matter just as much as raw model accuracy.

5.3 Limitations of the Study

Despite the successful development and evaluation of the proposed system, several limitations were found during implementation and evaluation. These mainly concern the scope of the study, the model implementation, and the evaluation method. They do not affect the feasibility of the system, but addressing them would further improve its performance, reliability, and practical use.

5.3.1 Absence of Model Fine-Tuning

The system is based on the pre-trained BiomedCLIP model and the Qwen2.5-1.5B-Instruct model, which is used in the inference-only mode without further fine-tuning on the NIH ChestX-ray14 dataset. This was selected to reduce computation and to demonstrate ability to produce disease predictions and to provide radiology-style disease reports without retraining the foundation model. The models were not optimized for this particular data set, however, this means that its predictions still rely on the pre-training data. Using the domain-specific chest X-ray datasets to fine-tune the models may yield more accurate predictions, more precise clinical descriptions, and better overall performance of the system.

5.3.2 Limited Evaluation Scale

Although the NIH ChestX-ray14 dataset contains a large collection of chest X-ray images covering various thoracic diseases, the prototype was evaluated using a representative subset for demonstration, functionality testing, response time analysis, and output consistency evaluation. Therefore, the system has not been thoroughly evaluated with the entire data and with all disease variations and imaging conditions. A larger evaluation with more images and a wider range of disease categories would give a fuller picture of the system's robustness, generalisation, and reliability.

5.3.3 Dependence of Report Generation on Disease Prediction

The report generation generated relies on the disease prediction from the vision-language model as its main input. This means that the accuracy and reliability of a report is directly related to the accuracy of the disease prediction. The report will also be wrong if the first prediction is wrong or incomplete, because the language model does not analyse the image of the chest radiograph alone. This would directly enhance the quality and

reliability of the reports generated, by improving the disease prediction aspect.

5.3.4 Simplified PACS-Style Implementation

The developed prototype is not a clinical Picture Archiving and Communication System, but is a simplified version of the PACS type environment for research and education purposes. It successfully combines image upload, chest X-ray visualisation, AI-assisted disease prediction, report generation, database management, study history, and PDF export in one workflow. But several functions common in commercial PACS solutions were outside the scope of this study, including integration with Hospital Information Systems (HIS), Radiology Information Systems (RIS), DICOM network communication, user authentication, and multi-user collaboration. The prototype need not be considered a working clinical PACS system, but rather a demonstration of the concept of integration with the workflow.

5.3.5 Lack of Clinical Validation

The evaluation in this study focused mainly on technical functionality, system performance, output consistency, and workflow integration. Although the system performed stably in these tests, the generated disease predictions and radiology-style reports have not been clinically validated by certified radiologists or healthcare institutions. The prototype is meant to serve as an AI-assisted decision-support tool for research and educational use, not a replacement for professional medical diagnosis. Clinical validation with healthcare professionals would be needed before the system could be considered suitable for real clinical use.

5.4 Future Work

The proposed system fulfilled the goals of this research, but there are some improvements for enhancing its capability and clear application. Based on the limitations identified above, two main directions are recommended for future research.

5.4.1 Fine-Tuning of Foundation Models

Future research should look into fine-tuning the pre-trained Vision-Language Model and Large Language Model using large-scale, domain-specific chest X-ray datasets. In this study, the system used the pre-trained foundation models in an inference-only setup without extra training. While this successfully showed that AI can be integrated into a PACS-style workflow, fine-tuning could improve disease prediction accuracy, raise the quality and consistency of the AI-generated reports, and help the models better capture domain-specific radiological features and medical terms. Evaluating the fine-tuned models on external datasets would also give a fuller picture of their robustness and generalisation across different clinical settings. In practical terms, fine-tuning the foundation models on large-scale, domain-specific chest X-ray datasets would directly improve diagnostic accuracy, which in a hospital setting means fewer missed or misclassified findings and more confidence for the reviewing radiologist.

5.4.2 Clinical Integration and Validation

Future work should focus on extending the prototype into a clinically usable decision-support system, through validation by healthcare professionals and integration with existing hospital infrastructure. Clinical evaluation involving certified radiologists would give useful feedback on the accuracy, reliability, and practical usefulness of the generated disease predictions and reports. The prototype could also be integrated with hospital systems such as PACS, HIS, and RIS using standard DICOM communication protocols,

which would allow automated retrieval, storage, and management of medical images while supporting a more realistic clinical workflow. These improvements would make the system more practical and give a stronger foundation for future AI-assisted medical imaging applications. In practical terms, integrating this system into a real clinical workflow could reduce a radiologist's daily workload by pre-generating draft reports for routine studies, letting specialists focus on complex or ambiguous cases while improving both throughput and diagnostic accuracy in a busy hospital department.

REFERENCES

- Arora, D., & Mehta, Y. (2014). Use of picture archiving and communication system for imaging of radiological films in cardiac surgical intensive care unit. *Journal of Anaesthesiology Clinical Pharmacology*, 30(3), 447–448. <https://doi.org/10.4103/0970-9185.137306>
- Demner-Fushman, D., Kohli, M. D., Rosenman, M. B., Shooshan, S. E., Rodriguez, L., Antani, S., Thoma, G. R., & McDonald, C. J. (2016). Preparing a collection of radiology examinations for distribution and retrieval. *Journal of the American Medical Informatics Association*, 23(2), 304–312.
- He, Z., Wong, A. N. N., & Yoo, J. S. (2025). Radiology report generation using automatic keyword adaptation, frequency-based multi-label classification and text-to-text large language models. *Computers in Biology and Medicine*, 196, 110625. <https://doi.org/10.1016/j.compbiomed.2025.110625>
- Hopstaken, R. M., Witbraad, T., van Engelshoven, J. M. A., & Dinant, G. J. (1996). Interobserver variation in the interpretation of chest radiographs for pneumonia in community-acquired lower respiratory tract infections. *European Journal of Radiology*, 29(1), 29–35. [https://doi.org/10.1016/0720-048X\(95\)00716-3](https://doi.org/10.1016/0720-048X(95)00716-3)
- Hu, D., Zhang, S., Liu, Q., Zhu, X., & Liu, B. (2025). Large language models in summarizing radiology report impressions for lung cancer in Chinese: Evaluation study. *Journal of Medical Internet Research*, 27, e65547. <https://doi.org/10.2196/65547>
- Huemann, Z., Tie, X., Hu, J., & Bradshaw, T. J. (2024). ConTEXTual Net: A multimodal vision-language model for segmentation of pneumothorax. *Journal of Imaging Informatics in Medicine*, 37, 1652–1663. <https://doi.org/10.1007/s10278-024-01051-8>
- Imaging the Chest. (2001). *Radiographic densities on chest radiographs*. <https://x-raylady.com/downloadfile/imagingthechest.pdf>

- Islam, M. R., Hossain, M. Z., Ahmed, M., & Samu, M. S. (2025). Vision-language models for automated chest X-ray interpretation: Leveraging ViT and GPT-2. *ArXiv Preprint ArXiv:2501.12356*. <https://doi.org/10.48550/arXiv.2501.12356>
- Joy, M. (2024). AI integration in PACS: Advancing medical imaging and healthcare delivery. *International Journal of Scientific and Advanced Technology*, 16(1), 2930. <https://doi.org/10.71097/IJSAT.v16.i1.2930>
- Lin, M., Holste, G., Wang, S., Zhou, Y., Wei, Y., Banerjee, I., Peng, Y., & others. (2025). CXR-LT 2024: A MICCAI challenge on long-tailed, multi-label, and zero-shot disease classification from chest X-ray. *ArXiv Preprint ArXiv:2506.07984*. <https://doi.org/10.48550/arXiv.2506.07984>
- Ng, C., Sun, L., & Tang, S. (2025). X-Ray-CoT: Interpretable chest X-ray diagnosis with vision-language models via chain-of-thought reasoning. *ArXiv Preprint ArXiv:2508.12455*. <https://doi.org/10.48550/arXiv.2508.12455>
- Nguyen, N. H., & others. (2021). Deployment and validation of an AI system for chest X-ray interpretation. *ArXiv Preprint ArXiv:2104.02256*. <https://arxiv.org/abs/2104.02256>
- Pham, T.-T., Brecheisen, J., Wu, C. C., Nguyen, H., Deng, Z., Adjero, D., Doretto, G., Choudhary, A., & Le, N. (2025). ItpCtrl-AI: End-to-end interpretable and controllable artificial intelligence by modeling radiologists' intentions. *Artificial Intelligence in Medicine*, 160, 103054. <https://www.sciencedirect.com/science/article/pii/S0933365724002963>
- Qwen Team. (2024). Qwen2.5 Technical Report. *ArXiv Preprint ArXiv:2412.15115*. <https://doi.org/10.48550/arXiv.2412.15115>
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 8748–8763. <https://doi.org/10.48550/arXiv.2103.00020>

- Ryu, J. S., Kang, H., Chu, Y., & Yang, S. (2025). Vision-language foundation models for medical imaging: A review of current practices and innovations. *Biomedical Engineering Letters*, 15, 809–830. <https://doi.org/10.1007/s13534-025-00484-6>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 618–626. <https://doi.org/10.1109/ICCV.2017.74>
- Stec, N., Arje, D., Moody, A. R., Krupinski, E. A., & Tyrrell, P. N. (2018). A systematic review of fatigue in radiology: Is it a problem? *American Journal of Roentgenology*, 210(4), 799–806. <https://doi.org/10.2214/AJR.17.18613>
- Tanno, R., Barrett, D. G. T., Sellergren, A., Ghaisas, S., Dathathri, S., See, A., Welbl, J., Lau, C., Tu, T., Azizi, S., Singhal, K., Schaekermann, M., May, R., Lee, R., Man, S., Mahdavi, S., Ahmed, Z., Matias, Y., Barral, J., ... Ktena, I. (2024). Collaboration between clinicians and vision–language models in radiology report generation. *Nature Medicine*, 30, 3302–3321. <https://doi.org/10.1038/s41591-024-03302-1>
- Thawakar, O., Shaker, A., Mullappilly, S. S., Cholakal, H., Anwer, R. M., Khan, S., Laaksonen, J., & Khan, F. S. (2024). XrayGPT: Chest radiographs summarization using large medical vision-language models. *Proceedings of the 23rd Workshop on Biomedical Language Processing*, 440–448. <https://aclanthology.org/2024.bionlp-1.35.pdf>
- Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2097–2106. <https://doi.org/10.1109/CVPR.2017.369>

- Wang, Y., Wang, Z., Zhu, D., Zhong, J., & Li, W. (2025). Governance-ready small language models for medical imaging: Prompting, abstention, and PACS integration. *ArXiv Preprint ArXiv:2508.13300*. <https://arxiv.org/abs/2508.13300>
- Yang, A., Yang, B., Hui, B., Zheng, B., Yu, B., Zhou, C., & others. (2024). Qwen2 Technical Report. *ArXiv Preprint ArXiv:2407.10671*. <https://doi.org/10.48550/arXiv.2407.10671>
- Zhang, S., Xu, Y., Usuyama, N., Xu, H., Bagga, J., Tinn, R., Preston, S., Rao, R., Wei, M., Valluri, N., Wong, C., Tupini, A., Wang, Y., Mazzola, M., Shukla, S., Liden, L., Gao, J., Crabtree, A., Piening, B., ... Poon, H. (2025). BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image–text pairs. *ArXiv Preprint ArXiv:2303.00915*. <https://doi.org/10.48550/arXiv.2303.00915>